Glasma Role in Jet Quenching Effects

José Soares Trindade Nunes dos Santos

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Supervisor(s): Dr. Liliana Apolinário
Dr. Pablo Guerrero Rodriguez

Examination Committee

Chairperson: Prof. Full Name 1
Supervisor: Prof. Full Name 2
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Month Year

Dedicated to someone special...

Declaration

I declare that this document is an original work of my own authorship and that it fulfills all the requirements of the Code of Conduct and Good Practices of the Universidade de Lisboa.

Acknowledgments

A few words about the university, financial support, research advisor, dissertation readers, faculty or other professors, lab mates, other friends and family...

Resumo

Inserir o resumo em Português aqui com um máximo de 250 palavras e acompanhado de 4 a 6 palavras-chave.

Palavras-chave: palavra-chave1, palavra-chave2, palavra-chave3,...

Abstract

Insert your abstract here with a maximum of 250 words, followed by 4 to 6 keywords.

Keywords: keyword1, keyword2, keyword3,...

Contents

List of Tables

List of Figures

Glossary

BFKL Balitsky–Fadin–Kuraev–Lipatov xviii, 11, 12, 16

BK Balitsky-Kovchegov 16

CGC Color Glass Condensate 13, 15, 17–19, 22, 24

CM Center of Mass 5

DGLAP Dokshitzer–Gribov–Lipatov–Altarelli–Parisi xviii, 11, 12

DIS Deep Inelastic Scattering 9

HERA Hadron–Electron Ring Accelerator 9

HIC Heavy Ion Collision 5–7, 13–16, 19, 24

IMF Infinite Momentum Frame 15, 16

JIMWLK Jalilian-Marian–Iancu–McLerran–Weigert–Leonidov–Kovner 19

LHC Large Hadron Collider 5, 6, 9

MV McLerran-Venugopalan 15, 16, 19

NNLO Next-to-Next-to-Leading-Order 10

PDF Parton Distribution Function 9–11

QCD Quantum Chromodynamics 2–6, 10, 11, 13, 15, 16

QED Quantum Electrodynamics 2, 3

QGP Quark-Gluon Plasma 5–8, 15

RHIC Relativistic Heavy Ion Collider 5, 6

Chapter 1

Introduction

1.1 Particle Physics and the Standard Model

The main objects of study of Particle Physics are the fundamental constituents of Nature, the particles, and the interactions between them, the forces.

Starting from the perspective of pre-mid-20th century physics, four fundamental forces between particles, observed from then-known physical phenomena, may be listed: Gravity, between particles with mass; Electrodynamics, between particles with electric charge; the Strong Force, responsible for keeping nucleons together in atomic nuclei; and the Weak Force, responsible for the radioactive decay of nuclei involving the conversion of a proton into a neutron or vice-versa. Around the same time, a suspected set of fundamental particles could comprise the proton, the neutron, the electron and the neutrino.

The Standard Model of Particle Physics aims to explain natural phenomena starting from a a set of fundamental particles and the mathematical framework, provided by Quantum Field Theories, to describe interactions between them. It was developed over the second half of the 20th century with contributions from many physicists, and is often referred to as the most successful modern physical theory, having produced a variety of theoretical predictions which have been independently confirmed through experiment. These include the prediction of several particles before their observation and the famously precise prediction of the anomalous magnetic moment of the electron, which agrees with experiment to roughly one part in 10^{10} [1]. However, there is also no shortage of physical phenomena currently unexplained by the Standard Model, such as gravity, baryon asymmetry (the prevalence of matter over antimatter in the Universe) and dark matter.

The set of fundamental particles in the Standard Model differs from the earlier suspected set. In the static quark model, the proton and the neutron are revealed to be composite particles, with nucleons being made of three quarks, of up- or down- flavor. Later, particles generated from the interactions between these quarks, as will be discussed later, were added to the composition of the nucleons. This new set of fundamental particles – the up-quark, the down-quark, the electron and the neutrino – is the first generation of fermions, which, roughly, are the particles involved in physical phenomena at our energy scale. At greater energy scales, two other generations come into play, each made up of four particles,

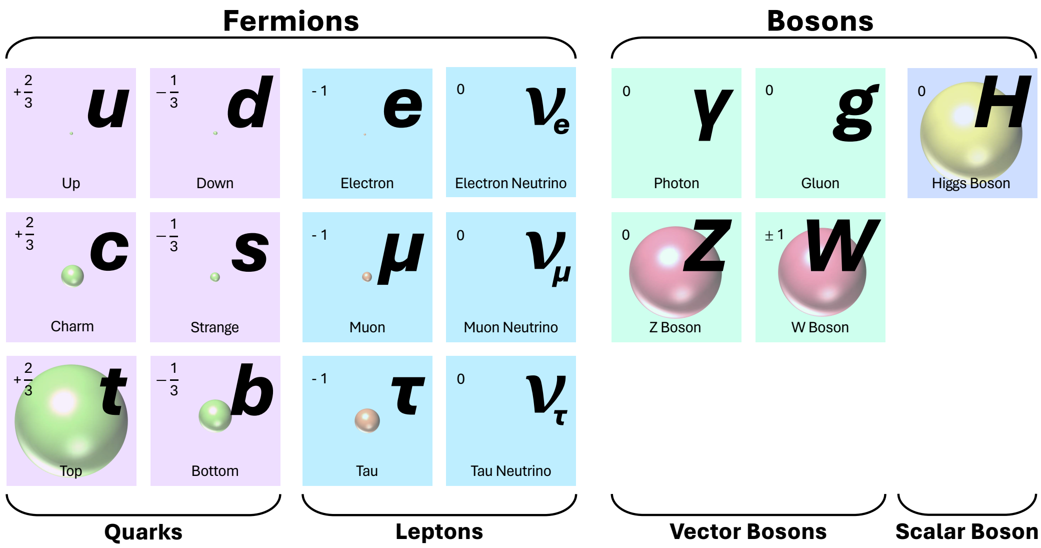


Figure 1.1: The elementary particles of the Standard Model. For each particle, its charge relative to the electron is shown and its mass is illustrated as proportional to the volume of a coloured sphere.

copies of the first four, differing only in mass. For each fermion, there also is an oppositely-charged copy, its antiparticle. Additionally, interactions between particles at a distance are always mediated by other particles, with each force having an associated mediator particle, called a vector boson: the photon for Electrodynamics, the W and Z bosons for the Weak Force, and the gluon for the Strong Force (Fig. 1.1).

The forces' relative strengths allow us to determine which force, out of the ones allowed in a system, is dominant: if the strong force has a strength of order 1, the electromagnetic force's strength is of order 10^{-3}, the weak force's strength is of order 10^{-8}, and gravity's strength is of order 10^{-37}. [2]

1.1.1 The Strong Force

In systems made up of quarks and gluons, which will be the subject of this work, the Strong Force is the dominant fundamental force.

Only particles with color charge experience the Strong Force – this includes only the quarks and the strong force's own mediator particle, the gluon. This is in contrast with Electrodynamics, where the mediator particle, the photon, does not carry electric charge. Color is actually a set of three charges: red charge, green charge and blue charge. All quarks and gluons have non-zero color charge, and all interactions between them – and with any other particles – conserve the total color charge over time. The allowed interactions between quarks and gluons are shown in Fig. 1.2.

The fact that the Strong Force is mediated by a massless boson between particles carrying its associated charge may suggest that it should be quantitatively treated and behave similarly to Electrodynamics. However, this not the case. The crucial difference between the Strong Force and Electrodynamics manifests in the fact that the gluon may interact with itself, while the photon may not. Quantum Chromodynamics (QCD), the Strong Force's quantum field theory in the Standard Model, is an SU(3) gauge theory, and the non-abelian nature of the SU(3) group is the origin for the gluon's self-interaction. U(1), the group of the gauge theory for Quantum Electrodynamics (QED), is an abelian group, due to which

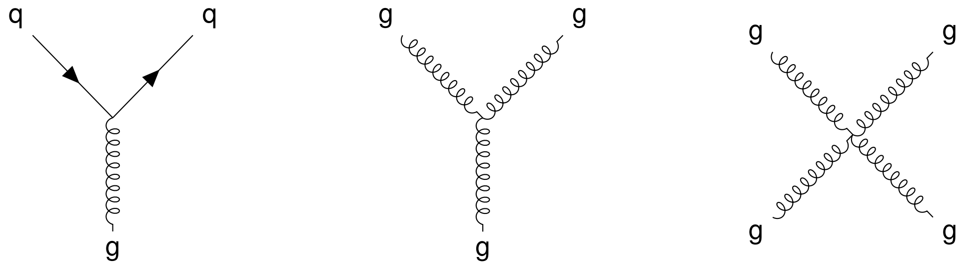


Figure 1.2: The interactions of the Strong Force. Solid, oriented lines represent quarks and "spring" lines represent gluons.

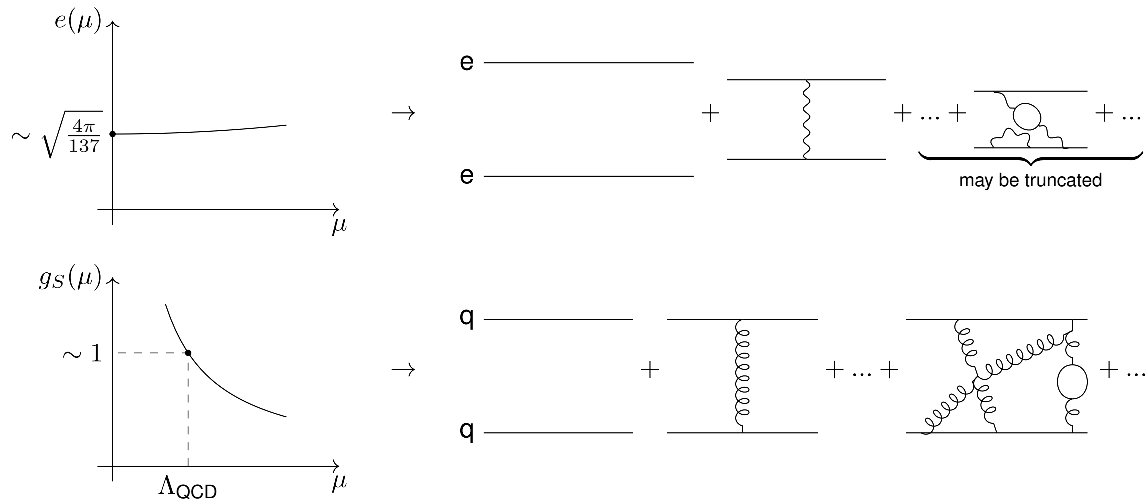


Figure 1.3: (Left) Rough dependence of the couplings of QED (top) and QCD (bottom) on the energy scale μ of a studied process. At low μ, $e(\mu)^2/4\pi$ approaches the value of the fine structure constant, $\alpha = e^2/4\pi \approx 1/137$. Below $\Lambda_{\text{QCD}} \approx 150\,\text{MeV}$, g_S is greater than one, disallowing the truncation of higher-order, more complex terms, the Perturbation Theory technique applicable for a wide range of energy scales in QED. (Right) Examples of the terms in the power series for the interaction between two electrons (top) and two quarks (bottom). For the quark pair, infinitely many diagrams with an increasing number of exchanged particles must be considered.

the photon does not interact with itself. This leads to the vastly different observed behaviors of QCD and QED. Once a Quantum Field Theory is renormalized, its strength, expressed by its coupling constant – usually, e for QED and g_S for QCD –, becomes dependent on the energy scale[1] of the phenomenon being described, μ, earning it the name of "running coupling". e acquires a relatively weak dependence on μ, and grows with it. Non-abelian gauge theories, on the other hand, may have their coupling constants decrease with μ; this is the case for g_S, which has a strong dependence on μ. Consequently, at low μ, it increases to values that no longer permit handling QCD via Perturbation Theory – the writing of a quantity as a power series in the coupling constant, and the truncation of that power series in order to allow its computation. Since the power of g_S associated with a process grows with the number of Strong Force vertices, this means that in order to calculate any result one would have to sum the contributions of an infinite number of increasingly complex diagrams, as illustrated in Fig. 1.3.

The case of low energies is equivalent to that of long distances. Color-charged particles separated

[1] Or, equivalently, its length or time scale.

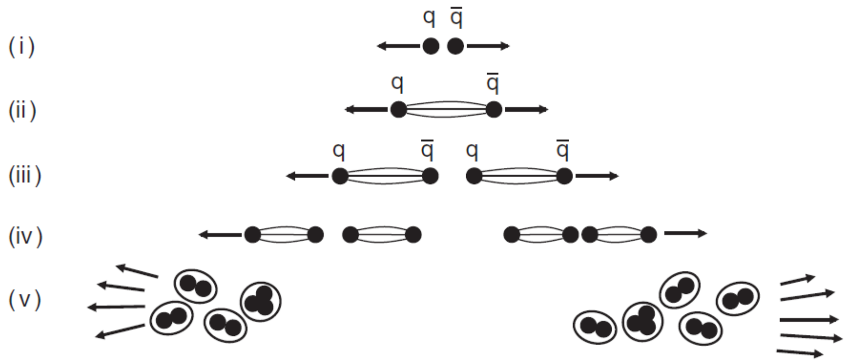


Figure 1.4: The generation of two jets from a separating $q\bar{q}$ pair. From [2].

over long distances are subject to the Strong Force with a large, non-perturbative coupling. As in Fig. 1.3, more complex processes are not necessarily less likely. Thus, gluons exchanged between any two color-charged particles moving away from one other are likely to exchange gluons between themselves and form quark-antiquark pairs ($q\overline{q}$). As such, the total energy stored in the fields between the distancing pair of particles increases until it becomes more energetically favorable for a ($q\overline{q}$) pair to remain between the two original particles, "screening" their charges and shortening the interaction distances. After this, if the original particles retain enough momentum to keep moving away from each other, the process repeats until only groups of particles in color singlets (with no overall color charge), such as mesons, protons or neutrons, remain – hadronization. This is the qualitative explanation for the fact that no isolated color-charged particles, like quarks and gluons, have been observed; this is called color confinement, for which an analytical explanation has yet to be found. Only composite particles, like hadrons, such as the proton and the neutron, have been detected, even though evidence for the existence of quarks and gluons exists. Conversely, at short distances or high energies, the coupling decreases logarithmically, leading to the QCD regime called asymptotic freedom, where quarks and gluons can exist isolated and where Perturbation Theory is applicable.

1.1.2 Jets

In Particle Physics experiments at colliders, a high energy quark or gluon is often produced which travels away from its birthing interaction vertex towards the surrounding particle detectors. Eventually, however, the distance between it and other color-charges particles grows until the process of hadronization takes over and a shower of colorless hadrons is generated. This shower of hadron lies in a "cone" around the original quark or gluon's momentum direction and is called a jet. It is the hadrons in this jet which actually reach the detectors, and several of its quantitative characteristics provide information on its founding particle. For example, the sum of the hadrons' four-momenta returns the founding particle's four-momentum. The process of jet formation from two receding quarks is illustrated in Fig. 1.4.

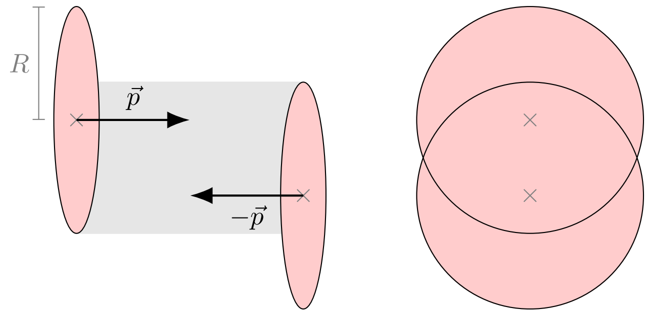


Figure 1.5: Non-central HIC in the CM frame – side (left) and transverse (right) views. Ions are represented as spheres which are length-contracted (not to scale) in the direction of their 3-momentum. Typical heavy ion radii (gold, lead) are of the order $R \approx 7\,\mathrm{fm}$. For non-central collisions, the overlap region is lenticular.

1.2 Heavy Ion Collisions

At collider experiments, beams of particles are accelerated and made to collide with each other, and the products of the collisions are analysed in order to gain insight into the interactions that took place.

When it comes to the study of the Strong Force, the interest is in gathering data relating to the interactions of quarks and gluons. In choosing what species of particles to accelerate and collide, it may be desirable for strong force-sensitive particles to only be products of the collision, or to be present from the beginning, forming the colliding beams. In the second case, only color-neutral composite particles are available due to color confinement, and the most readily available ones are the nuclei of atoms, the simplest of which is the proton, p; heavier ones, like gold and lead, are dubbed heavy ions, and are denoted by A. The focus of this work will be on Heavy Ion Collisions (HICs), also know as AA Collisions, as they produce the appropriate conditions for the formation of the Quark-Gluon Plasma (QGP), which will be discussed further ahead.

HICs are performed at CERN's Large Hadron Collider (LHC) in Geneva and at Brookhaven National Laboratory's Relativistic Heavy Ion Collider (RHIC) in Upton, New York. At these colliders, beams of hadrons are accelerated in opposite directions to close to the speed of light. At the highest energies attainable at the LHC, the Lorentz factor corresponding to the hadrons' velocities is of around $\gamma \approx 2500$; as such, they experience length contraction of the same factor and become very thin disks, as opposed to approximate spheres. The Center of Mass (CM) frame energy of these collisions is denoted by $\sqrt{s}$. [3] Fig. 1.5 represents a heavy ion collision in the CM frame.

The beams are made to collide at points of the accelerator equipped with detectors, which are mounted cylindrically around the accelerator tube. Consequently, only particles produced in the collision with enough transverse momentum reach the detectors, while other particles continue along the accelerator. As said before, no free quarks or gluons are detected, and any high energy partons which may have been produced hadronise and form jets before reaching the detectors.

As for the region of the collision, the high density of energies generated allows reaching exotic states of quark-gluon matter very distinct from the familiar ones. Fig. 1.6 presents a possible phase diagram for QCD matter. Its state variables are temperature, T, which is related to energy density in volume,

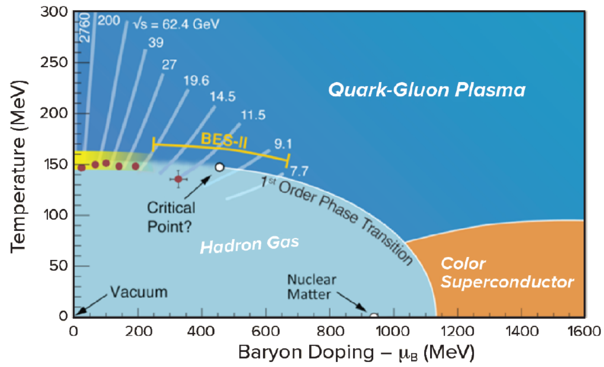


Figure 1.6: A possible QCD phase diagram. Also shown are lines representing the path drawn by cooling QGP droplets from HIC at different $\sqrt{s}$ and expected paths from the then-upcoming phase II of RHIC's Beam Energy Scan. From [3].

and baryon number chemical potential, μ_B, which translates into the excess in number of quarks over antiquarks.

Familiar nuclear matter is present in the diagram within the hadron gas phase at low temperatures and higher μ_B, as protons and neutrons contain a surplus of quarks over antiquarks. Near the vertical axis ($\mu_B \approx 0$, a good approximation for some of the matter produced in HICs at RHIC and the LHC), increasing the temperature of the hadron gas leads to a smooth transition into the QGP, a strongly-coupled fluid-like state of matter made of unconfined quarks and gluons, and the subject of the next section. Continuing upwards along the axis, although not represented in the phase diagram, another smooth transition into a gas-like, rather than fluid-like, QGP is predicted. Away from the vertical axis, it is posited that, upwards of some critical value of μ_B, the mentioned transition from a hadron gas into the QGP becomes a first order phase transition rather than a smooth one. Searches for this critical point can be made by studying the cooling of droplets of QGP formed in HICs at lower and lower energies (see different paths drawn by matter for different $\sqrt{s}$ values). It is further posited that, for great enough μ_B, a color superconductor phase is reached. [3]

1.2.1 The Quark-Gluon Plasma

Studying the properties of the QGP is of great interest. It was the Universe's first form of complex matter, and can also be thought of as the simplest form of complex matter, since it is a fluid whose constituents interact directly according to the rules of QCD (in the perturbative regime); these are fundamental, rather than effective, interactions, so the study of the QGP is of great importance for the understanding of QCD.

The QGP is formed in HICs as the two nuclei move away from each other post-encounter. During the collision, the constituent partons (quarks and gluons) of both nuclei interact in the overlap region via mostly soft (low) momentum exchanges[2] and some color exchanges. As they move away post-collision, partons from different nuclei exchange gluons, since there was exchange of color between the nuclei. These gluons decay into other gluons and $q\bar{q}$ pairs in the middle region. The medium created between

[2]Some hard (high momentum exchange) interactions do occur, which allow for some partons to acquire high transverse momentum. These will serve as important probes of the QGP as will be seen further ahead.

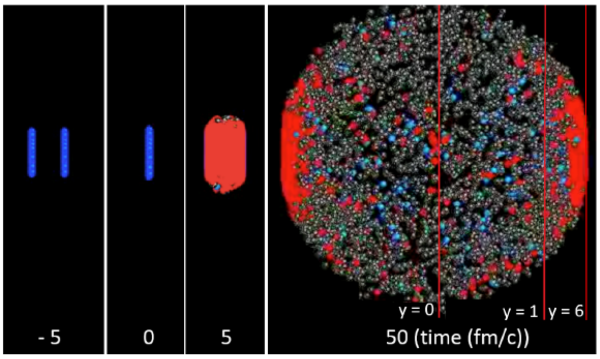


Figure 1.7: Snapshots from an animation representing a $\sqrt{s} = 2.76\,\mathrm{TeV}$ lead-lead central collision. Hadrons are represented by blue and grey spheres and QGP is shown in red. In the final frame, at time 50 fm/c, lines over particles with the same rapidity, y, a function of their distance to the transverse plane containing the center of the collision, are shown. From [3].

the separating nuclei is energetic enough not to hadronize but not enough for the constituent partons to be free from each other; they are instead strongly coupled, forming a ”soup” that is well described as a relativistic hydrodynamic fluid by the time the nuclei are about $2\,\mathrm{fm}$ apart, at time $1\,\mathrm{fm/c}$ post-collision. It is characterised by a viscosity to entropy density ratio $\eta/s \approx 1/4\pi$, in units with $\hbar = k_B = 1$, which is the smallest of any known fluid. New QGP continuously forms at later and later times in the wake of the receding nuclei, because it originates from increasingly fast partons which suffer greater time dilation. Once formed, each volume element of QGP expands and cools, driven by the initial pressure, until it reaches a low enough energy density which forces it to hadronize. Part of the ensuing cloud of hadrons then ends up with a momentum oriented towards the surrounding detectors. [3] Fig. 1.7 shows snapshots from an animation representing a central HIC.

The QGP formed at these collisions is too small and short-lived to be observed directly, so its mentioned properties have been deduced indirectly from the available observables. These include: 1) The number of produced charged particles and its dependence on $\sqrt{s}$ and the species of nuclei used. 2) The azimuthal distribution of energy deposited in the detectors being anisotropic, which suggests a fluid-like, low viscosity, nature for the QGP[3]. 3) The production rates of (charm containing) J/ψ and (bottom containing) Υ mesons, which suggest they traverse a medium where partons are unconfined. 4) Lastly, the metrics of hard probes – partons with high transverse momentum, p_T, originating from rare hard scatterings at the beginning of the collision – which traverse the QGP, losing energy to it in a phenomenon known as jet quenching and creating a wake. [3]

For the present work, the focus will be on this last avenue for gaining knowledge on the QGP. The existence of the QGP may be inferred from data on the jets founded by the mentioned high-p_T partons. If there were no QGP, a HIC could be thought of as equivalent to a certain number pp collisions[4], N_{coll}, determined by the centrality of the HIC. This approach works well when comparing the number of high-p_T photons and Z bosons, which are *not* Strong Force-sensitive, produced in HICs *vs.* pp collisions. However, performing the same comparison regarding high-p_T jets returns that the number and energy

[3]If the QGP were gas-like, azimuthal anisotropies arising from the non-uniform distribution of partons in the nuclei and the lenticular shape of the overlap region would fade away as it expands.

[4]Collisions between two protons.

of high-p_T jets is lower than expected, suggesting quenching by a medium. Furthermore, the amount of energy lost (around $10\,\mathrm{GeV}$), given the small length scales of the medium, suggests an enormous stopping power, which points to the strongly-coupled nature of the QGP.

1.2.2 The Pre-Hydrodynamic Phase

The above discussion has left out concerns about the behavior of the medium *before* it becomes hydrodynamic and reaches equilibrium. If one is hoping to extract information about the QGP from probes which traversed it but may have been formed prior to its hydrodinamization, then it is important to assess how they may be affected by the preceding medium, and by how this preceding medium evolves into the initial conditions of the QGP.

This will be the concern of the present work. [Summary of what will be done.]

$\times$

Chapter 2

2.1 A Picture of Nuclear Matter

Probing hadrons at high energies in scattering experiments has allowed exploring the internal structure of nucleons. Data from both fixed-target and collider (Hadron–Electron Ring Accelerator (HERA), Tevatron, LHC) experiments has been compiled in order to obtain pictures of the gluon and quark content of hadrons. One may refer collectively to quarks and gluons, when they are the constituents of a hadron, as partons.

At HERA, electrons (or positrons) and protons were accelerated to collide at center of mass energies of around $\sqrt{s} = 300\,\mathrm{GeV}$. In some of these collisions, the momentum transfer Q^μ from the electron to the proton via a virtual photon was large enough ($Q^2 > 200\,\mathrm{GeV}^2$) to disintegrate the proton into a shower of hadrons $(e + p \to e + X)$; these are called Deep Inelastic Scattering (DIS) processes, whose cross-sections are valuable in unveiling the proton's parton content. The greater the virtual photon's energy, the greater its ability to resolve structures of smaller scale than the proton radius and with shorter characteristic timescales, since the spatial and temporal resolutions of these experiments are roughly proportional to $1/Q$ or $1/\sqrt{s}$. DIS processes have also been studied for this purpose using muons and neutrinos as probes, and neutrons as targets – each probe selects for different processes of interaction with the hadron, which allow assessing the abundance of different, specific species of partons[1]; using the neutron as a target also provides insight into the internal structure of the proton due to their similarity: in the static quark model, the quark compositions of the proton and the neutron are uud and ddu. At the Tevatron and LHC colliders, data from $p\overline{p}$ and pp collisions, respectively, has contributed to the study of the gluon content of protons, specifically. [4][5] Unlike at HERA, in these experiments the two projectiles may exchange a virtual color-charged particle, which allows for processes useful in assessing the abundance of gluons without interference from the quark content. [5, Table 18.3]

The contributions from these many experiments have permitted the quantitative assessment of the proton's parton content via the measurement of its Parton Distribution Functions (PDFs). These translate the abundance, inside the proton, of a given parton species carrying a certain fraction of its momentum (in a frame of reference where the proton's momentum is large). A species q's PDF, $q(x)$, is defined such that the number of q partons carrying a fraction of the proton's momentum between x and $x + \Delta x$,

[1] For example, the use of neutrinos as probes permits scanning for $\nu + N \to \mu^- + X$ (with N a nucleon) events, where it is guaranteed that a W^+ boson was exchanged between the electron and a negatively charged quark or antiquark.

in a frame where the proton's momentum is much greater than its mass, is given by $q(x)\Delta x$. These functions actually also depend on the energy scale of the process used to probe the proton, Q^2, since a greater exchange of energy allows probing finer features of the proton, increasing its PDFs at small x, for example. [4]

Fig. 2.1 shows a recent result for the proton's PDFs. For $0.1 \lesssim x < 1$, the "valence" quarks u ($\mathbf{u}_V$) and d ($\mathbf{d}_V$) dominate, and in the relative proportion expected from the static quark model, uud. However, for $x \lesssim 0.02$, they are overtaken by a "sea" of quarks and antiquarks. These may be thought of as $q\overline{q}$ pairs arising from interactions between the valence quarks. Therefore, any "sea" flavor's PDF is identical to its antiquark's PDF. The "sea" includes new flavors (s, c and b) and u and d quarks which are as abundant as the $\overline{u}$ and $\overline{d}$ antiquarks; to distinguish them from their valence counterparts, their PDFs are written $\overline{\mathbf{u}}$ and $\overline{\mathbf{d}}$. For small x, however, the most abundant parton is, by far, the gluon ($\mathbf{g}$).

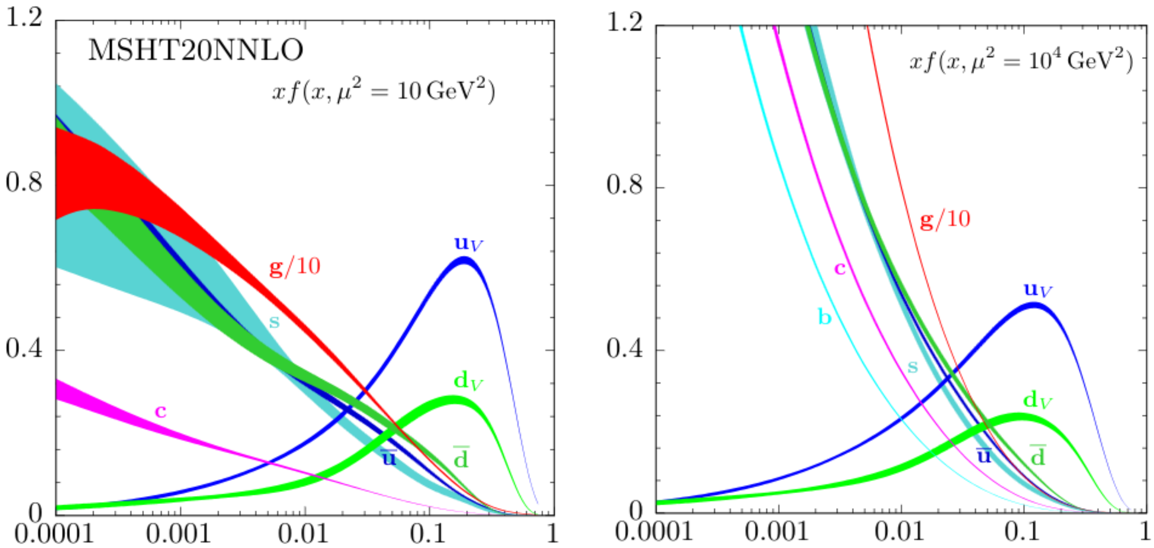


Figure 2.1: The proton's PDFs multiplied by x calculated at Next-to-Next-to-Leading-Order (NNLO) for the case of unpolarized beams at the energy scales $\mu^2 = 10\,\mathrm{GeV}^2$ (left) and $\mu^2 = 10^4\mathrm{GeV}^2$ (right). The gluon's PDF appears scaled down by a factor of 10. [5] Originally from and calculated by the MSHT Group in 2020. [6]

2.2 Gluon Saturation

The dominance by the gluon over the proton's parton content at low-x is predicted by QCD. Two avenues for reaching this prediction will be explored in this section: first, using the splitting function of the gluon and, second, using the QCD evolution equations. $\times$

The splitting function of non-Abelian gauge bosons such as the gluon is a well know QCD result. It returns the probability of emission by a parton of a gluon with relative transverse momentum $\mathbf{k}_\perp$ and a fraction x of its longitudinal momentum. In the soft limit, at lowest order in α_S and for small x, it reads [7]:

$$\mathrm{d}P \propto \frac{\alpha_S}{\pi^2}\frac{\mathrm{d}^2\mathbf{k}_\perp}{|\mathbf{k}_\perp|^2}\frac{\mathrm{d}x}{x} \tag{2.1}$$

The probability then contains colinear ($|\mathbf{k}|_\perp \to 0$) and soft ($x \to 0$) logarithmic singularities. The singularity as $x \to 0$ suggests a picture of abundance of small-x gluons within all hadrons, which grows uncapped as partons emit small-x gluons, which themselves emit smaller-x gluons or produce quark-antiquark pairs which also serve as gluon emitters. However, gluon densities may not explode as $x \to 0$, otherwise unitarity would be violated. [8]

Below a certain value of x, non-linear dynamics not considered so far should take over and tame the divergence, *saturating* the low-x gluon density in the hadron. The onset of saturation is controlled by the saturation scale, $Q_s(x)$, a transverse momentum scale below which non-linear effects become relevant. [8]

One interpretation of these non-linear effects is the onset, above certain gluon densities, of gluon recombination. Above that threshold, the probability of emission of a new gluon becomes comparable to the probability of recombination of gluons, slowing the growth of their density to rates compatible with conservation of unitarity (Fig. 2.2).

Figure 2.2: The onset of gluon recombination. Above a certain gluon density, gluon recombination (red dot) becomes comparatively likely to gluon emission (the pink and blue ladders of gluons). Adapted from [8].

This uncapped growth of gluon densities in the proton is also quantitatively predicted by the evolution equations of QCD. These equations provide a picture of how the parton content of the proton should evolve with x and with the energy scale, μ.

The DGLAP differential equations describe how PDFs should evolve with μ at fixed x. They predict that all partons' PDFs at low-x should grow when μ is increased, as in Fig. 2.1. [9][2] The PDFs of the valence quarks, specifically, grow at low-x and shrink at high-x when μ is increased, effectively moving them back along the x axis. This reflects the fact that increasing the energy of the probing process allows resolving lower energy partons in the proton – low-x gluons emitted by a valence quarks may be distinguishable at a higher μ but seen as part of the valence quark at a lower μ (Fig. 2.3).

The BFKL differential equations, on the other hand, describe how gluon density in momentum space, ϕ, should evolve with x at fixed μ. It is a linear equation, and predicts a divergence of the gluon density as $x \to 0$; more specifically, it predicts an exponential growth of ϕ with rapidity, $y \equiv \ln(1/x)$. The equation

[2] In this book the DGLAP equations are referred to as the Altarelli-Parisi equations.

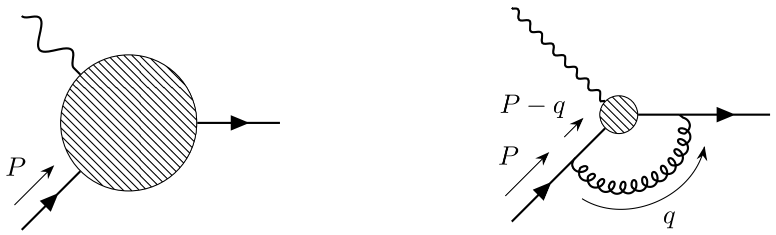


Figure 2.3: (Left) A quark being resolved as having momentum P by a low-energy photon. (Right) The same quark becoming distinguishable from a gluon it emitted when probed by a high-energy photon, which resolves it as having momentum $P - q$. Adapted from [2].

does not contain any terms reflecting interference between gluon emitters, as it assumes the hadron to be a dilute parton system. However, making the probability of emission of a new gluon dependent on the present gluon density by adding non-linear terms to the BFKL equations allows for saturation to be reached when gluon generation and gluon recombination balance out. These new non-linear terms have coefficients that are inverse powers of energy scales, $(1/Q)^n$; they control when the new terms become comparable to the old linear terms, and thus act as saturation scales, Q_s. [8]

Figure 2.4 illustrates the evolution of the parton content of the proton when its characterizing functions are evolved via the DGLAP and the BFKL equations. The upper left corner of this plot corresponds to the saturation region, where non-linear effects like recombination are thought to take over, otherwise gluon density growth as $x \to 0$ would violate unitarity.

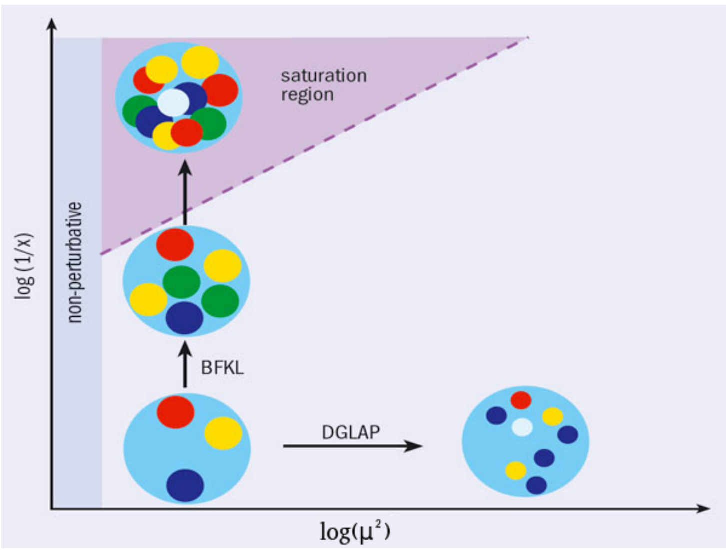


Figure 2.4: An energy scale, μ, vs. x plot illustrating the evolution of the parton content of the proton via the DGLAP and BFKL equations. Moving left to right via DGLAP, parton populations increase but the circles representing them get smaller, reflecting the decreasing characteristic length scale of the probing process ($\sim 1/\mu$). Moving bottom to top via BFKL, parton populations also increase but the partons' size remains the same; the region where there are enough partons to force the circles to overlap is used as an analog for the region of this phase space where the proton should become saturated – an area of interest for exploration in collider experiments. Adapted from [10].

The proton's (and, therefore, the heavy ion's) content may thus be thought of as dominated by its saturated low-x gluon population. Over the next sections, a formalism for handling systems like this, the

Color Glass Condensate (CGC), will be introduced, preceded by a review of the QCD Lagrangian and the definition of a proper set of coordinates in the context of a HIC.

2.3 The QCD Lagrangian

In studying a system where strong interactions are dominant, one turns to the QCD Lagrangian, which encodes those interactions. It is a functional of the quark fields, ψ, which are 4-component Dirac spinors (with $\overline{\psi} = \psi^\dagger\gamma^0$ being the quark field's Dirac adjoint), and of the gluon fields (also called color fields), A^a_μ, which are real-valued four-vectors; both are functions of spacetime. For a theory with N_c colors and N_f flavors of quarks each with mass m_f, the Lagrangian is

$$\mathcal{L}_{\text{QCD}} = \sum_{f=1}^{N_f} \overline{\psi}_f \left(i\gamma^\mu D_\mu - m_f\right)\psi_f - \frac{1}{2}\,\mathrm{Tr}_c\left[F_{\mu\nu}F^{\mu\nu}\right] \tag{2.2}$$

where D_μ, the covariant derivative, is given by:

$$D_\mu = \partial_\mu + igA_\mu, \quad \text{where } A_\mu = A^a_\mu T_a \tag{2.3}$$

and $F^{\mu\nu}$, the gluon field strength tensor, is given by:

$$\begin{aligned} F_{\mu\nu} &\equiv F^a_{\mu\nu}T_a \\ F^a_{\mu\nu} &= \partial_\mu A^a_\nu - \partial_\nu A^a_\mu - gf^{abc}A^b_\mu A^c_\nu \end{aligned} \tag{2.4}$$

where g is the Strong Force's coupling constant; T^a are the SU(N_c) generators in the fundamental representation[3], and f^{abc} are the SU(N_c) structure constants. Tr_c denotes a color-trace, that is, a trace in color space, the space of the T_a matrices' indices.

Writing the Lagrangian with all indices explicit,

$$\mathcal{L}_{\text{QCD}} = \sum_{f=1}^{N_f} \left(\psi^i_{\alpha f}\right)^* \left(\gamma^0\right)_{\alpha\beta} \left(i\left(\gamma^\mu\right)_{\beta\gamma}\left(\delta^{ij}\partial_\mu - igA^a_\mu\left(T^a\right)^{ij}\right) - \delta_{\beta\gamma}\delta^{ij}m_f\right)\psi^j_{\gamma f} - \frac{1}{4}F^a_{\mu\nu}F^{\mu\nu}_a \tag{2.5}$$

where $\alpha, \beta, \gamma = 1, 2, 3, 4$ are Dirac indices, $i, j = 1, ..., N_c$ are color indices in the fundamental representation, $a = 1, ..., {N_c}^2 - 1$ are color indices in the adjoint representation, $f = 1, ..., N_f$ are flavor indices and $\mu, \nu = 0, 1, 2, 3$ are spacetime indices when using Minkowski coordinates.

Given the presented QCD Lagrangian, the corresponding Euler-Lagrange equations relating to the gluon fields A are called the Yang-Mills equations, which can be thought of as the QCD equivalent to Classical Electrodynamics' Maxwell's equations. In the absence of quark fields ($\psi = 0$), they read:

$$\left[D_\mu, F^{\mu\nu}\right] = 0 \tag{2.6}$$

[3]Normalized such that $\mathrm{Tr}_c[T^a, T^b] = \delta^{ab}/2$

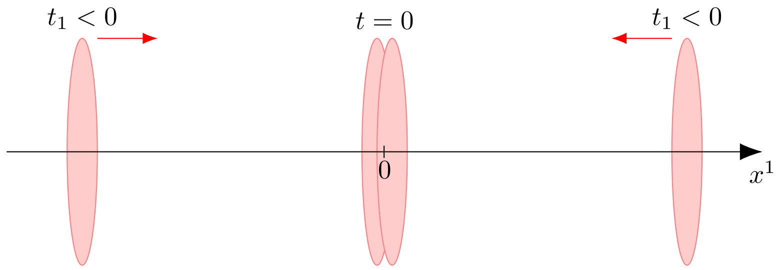


Figure 2.5: Two heavy nuclei, Lorentz-contracted into thin disks, travel along the x^1 axis in opposite directions until they collide at $t = 0, x^1 = 0$.

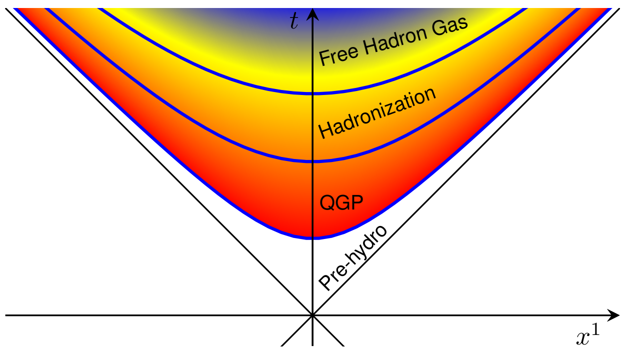


Figure 2.6: A heavy-ion collision represented in the x^1Ox^0 plane. Both nuclei travel along the light-cone centered around the collision at coordinates $(0, 0)$. The several stages of the post-collision inter-nuclear medium are delimited (not to scale). Based on the figure from [11].

2.4 A Set of Coordinates

In a HIC, as previously introduced, two heavy nuclei travel in opposite directions at near the speed of light before crashing into each other. For the purposes of this work, both nuclei will be taken to travel at the speed of light; because of this, they suffer length contraction along the direction of their trajectories, becoming infinitely thin/planar discs. Furthermore, their collision is taken to be central, that is, the trajectories of their centers are perfectly aligned.

In the laboratory/center of mass frame, the x^1 axis is placed along the trajectories of the centers of the nuclei. The nucleus moving in the positive x^1 direction is identified as nucleus 1, and the one moving in the negative x^1 direction is identified as nucleus 2. The nuclei collide at $x^1 = 0$, at instant $t = 0$ (Fig. 2.5). This system is invariant under boosts along the x^1 direction.

Prior to the collision, each nucleus's position four-vector, $x^\mu = (x^0, x^1, x^2, x^3)$, in Minkowski coordinates, is:

$$\text{Nucleus 1}: \quad (t, t, 0, 0)$$
$$\text{Nucleus 2}: \quad (t, -t, 0, 0)$$

These trajectories are well-illustrated in the x^1Ox^0 plane (Fig. 2.6).

After the collision, many of the nuclei's constituent particles carry on in the direction they were moving at the speed of light, and the pre-hydrodynamic medium which will be studied in this work forms in between the now receding discs. Fig. 2.6 suggests the usefulness of a coordinate system where the axes lie along the nuclei's path in the x^1Ox^0 plane. In the Light-Cone coordinate system (see Appendix B.1),

$$x^+ = \frac{x^0 + x^1}{\sqrt{2}} \tag{2.7a}$$

$$x^- = \frac{x^0 - x^1}{\sqrt{2}} \tag{2.7b}$$

In this coordinate system, the nuclei's position four-vectors, $x^\mu = (x^+, x^-, x^2, x^3)$, are:

$$\text{Nucleus 1}: \quad (\sqrt{2}t, 0, 0, 0)$$

$$\text{Nucleus 2}: \quad (0, \sqrt{2}t, 0, 0)$$

and the collision coordinates remain at $(x^+ = 0, x^- = 0)$.

In order to fulfill this work's objectives, it will be necessary to calculate the gluon fields generated in the space between the receding nuclei, making use of the classical Yang-Mills equations. The fields will be determined from the the static distributions of color charge of the receding nuclei.

Move the introduction to dEdx and $\hat{q}$ to before this? say they are the result of the color fields acting on the quark probe?

2.5 The Color Glass Condensate and The McLerran-Venugopalan Model

The Color Glass Condensate (CGC) is an effective theory of QCD for high-energy scattering applied to describe the aforementioned saturated gluons [7], which evade standard perturbation techniques. It is one of the approaches adopted to study initial state effects in HICs; that is, the properties of the system prior to the formation of the QGP, and their impact on final-state observables. It is used, for example, in the modeling of the incoming nuclei and of the non-equilibrated medium that leads to the QGP. In this regard, it has stood out due to its practical applicability and phenomenological success. [8]

The effective theory hinges on the partition of the parton content of a hadron, through approximations, into static color charges and the saturated gluons (represented by dynamic classical gluon fields) which they generate via the processes described in Section 2.2. [7] In this case, the hadron in question is a heavy nucleus, which is made up of neutrons and protons whose contents were previously discussed. This partition is done in the Infinite Momentum Frame (IMF), where the nucleus is boosted along its direction of motion in order to make all of its constituents' momenta approximately collinear to its own.

The founding iteration of this formalism was introduced in [12] and is known today as the McLerran-Venugopalan (MV) model. Together with other iterations, which incorporate quantum corrections to the classical theory, they form the corpus of the CGC framework.

2.5.1 Classical saturated gluon fields

YET UNJUSTIFIED: why $N \sim 1/g^2$; why g is small; well sourced?

First, the treatment of the saturated gluons as classical fields will be justified. Reaching the saturation regime implies that the non-linear terms of the evolution equation have become non-negligible[4], as discussed in Section 2.2. For this to occur, gluon densities should be of the order $\sim 1/\alpha \sim 1/g^2$. When taking the gluon density as a proxy for the gluon field two-point function, $\langle AA \rangle$, this allows estimating the average value of the gluon fields as $\sim 1/g$, which does not allow their treatment through standard perturbative techniques[5]. [8] However, it opens the possibility of their treatment as classical fields.

Comparing the occupation number of the saturated gluon fields, which is itself a proxy for the gluon field two-point function, and the commutator of their creation and annihilation operators, a and $a^\dagger$, [8] in the regime of perturbative QCD,

$$N = a^\dagger a \sim \frac{1}{g^2} \quad \gg \quad [a^\dagger, a] \sim 1 \tag{2.8}$$

it is possible to state that quantum corrections to the classical solution, where the fields commute, are negligible. Thus, it is reasonable to treat the low-x gluon fields as classical fields. This is the basis for the development of the McLerran-Venugopalan (MV) model.

2.5.2 Partition of the nucleus's parton content

In order to analitically validate the partition of the parton content of the nucleus into a static and a dynamic part in a HIC, it is necessary to move to the Infinite Momentum Frame (IMF). In the IMF, the nucleus is boosted along its direction of motion so as to make the momenta of its constituent partons approximately collinear to its own. In this frame, considering a nucleus moving in the positive x^1 direction with large total longitudinal momentum P much greater than its mass, m_A, the four-momenta of the nucleus and of one of its constituent partons in Minkowski coordinates are: [4]

$$\text{Nucleus: } P^\mu = (P, P, 0, 0) \qquad\qquad \text{Parton } i\text{: } p^\mu = (x_i P, x_i P, \sim 0, \sim 0) \tag{2.9a}$$

where $0 < x_i < 1$ is the fraction of the nuclear longitudinal momentum carried by the i-th parton; it was further assumed that the boost applied was such that $x_i P \gg m_i$, with m_i being the i-th parton's mass. In light-cone coordinates,

$$\text{Nucleus: } P^\mu = (P^+, 0, 0, 0) \qquad\qquad \text{Parton } i\text{: } p^\mu = (x_i P^+, \sim 0, \sim 0, \sim 0) \tag{2.9b}$$

where $P^+ = \sqrt{2}P$. It becomes clear that x_i is also the i-th parton's fraction of the nuclear light-cone $+$-momentum.

[4]Here, specifically, the Balitsky-Kovchegov (BK) equation at leading order is considered, where it serves as a non-linear extension of the BFKL evolution equation. [8]

[5]For example, in calculating the matrix element $\mathcal{M}$ for a QCD process using a series expansion in powers of g, all terms are proportional to $gA \sim \mathcal{O}(1)$, which does not allow truncating the series in order to estimate $\mathcal{M}$ – all orders need to be summed.

The uncertainty principle can be used to estimate the approximate extent of each parton over the x^- and x^+ coordinates: [8, 13]

$$\Delta x^- \propto \frac{1}{p^+} = \frac{1}{x_i P^+} \tag{2.10a}$$

$$\Delta x^+ \propto \frac{1}{p^-} = \frac{2p^+}{(p^0)^2 - (p^1)^2} = \frac{2p^+}{(\mathbf{p}_\perp)^2 + {m_i}^2} \approx \frac{2x_i P^+}{(\mathbf{p}_\perp)^2} \tag{2.10b}$$

where $\mathbf{p}_\perp$ is the i-th parton's transverse momentum, which was taken to be negligible in Eqs. (2.9) when compared to its longitudinal momentum.

Two useful conclusions about the x^+, x^- scales of high-x and low-x partons may be extracted from the uncertainty relations:

1) high-x partons are highly localized in the x^- direction. That is, they occupy a very thin "slice" of space in the x^- direction.

2) high-x partons are more delocalized in the x^+ direction than their low-x counterparts. That is, their typical x^+ scale is greater than the low-x partons'. Equivalently, the fluctuations of the low-x parton fields are much shorter along the x^+ direction than for the high-x parton fields.

It is from the second point that a hierarchy arises, central to the CGC formalism, of partons separated according to their x-values. A $+$-momentum scale, Λ^+, is introduced to separate the high-x, or valence, partons and the low-x partons. From the perspective of the low-x partons, the valence partons have very slow dynamics in x^+. In the limit of this argument, they are *static* in x^+; for particles traveling at the speed of light, being static in x^+ is equivalent to being static in t. As such, in the CGC, valence partons become constant color charges, which serve as sources for the *dynamic* low-x parton fields. As was introduced in sections 2.1 and 2.2, the partonic content of protons, and hence of the nucleus, is dominated by gluons at low-x. Therefore, the dynamic fields are taken to be purely gluon fields.

The parton content of the nucleus is thus divided into the group of high $+$-momentum ($p^+ > \Lambda^+$) static color charges and the group of low $+$-momentum ($p^+ < \Lambda^+$) dynamic classical gluon fields which they generate. The static color charges may themselves be treated as classical [8, 12], and are expressed in the formalism as a classical color current four-vector, $J^\mu \equiv J^\mu_a T^a$. This current is eikonally define coupled to the classical gluon fields; that is, the static charges suffer negligible recoil from their interactions with the gluon fields, when compared to their longitudinal momentum. [7] This is expressed by inserting the color current into the Lagrangian via the term $A^a_\mu J^\mu_a$. Since, now, the formalism does not include quantum quark fields, Eq. (2.5) becomes

$$\mathcal{L}'_{\text{QCD}} = -\frac{1}{4} F^a_{\mu\nu} F^{\mu\nu}_a + A^a_\mu J^\mu_a \tag{2.11}$$

where $J^\mu \equiv J^\mu_a T^a$.

Consequently, the corresponding Euler-Lagrange equations for the gluon fields A^μ become:

$$[D_\mu, F^{\mu\nu}] = J^\nu \tag{2.12}$$

which, provided an expression for the external color current, may be solved for the classical gluon fields.

Seeing as J^μ describes partons traveling at near the speed of light in the x^1 direction, then, in light-cone coordinates, its only non-vanishing component is J^+ (Eq. (2.13)). Additionally, it will directly depend on the number color charge distribution of the valence partons, $\rho(x) \equiv \rho_a(x)T^a$ (Eq. (2.14)).

$$J^\mu = \delta^\mu_+ J^+ \tag{2.13}$$

$$J^+(x^\mu) = g\rho(x^\mu) \tag{2.14}$$

From the previous discussion on the static nature of the color charges, ρ's dependence on x^+ can be dropped. Furthermore, due to the first conclusion drawn from the uncertainty relations (Eqs. 2.10), the color charge distribution function describing the placement of the static valence partons is very thin in the x^- direction. The valence color current becomes, in light-cone coordinates:

$$J^\mu(x^-, \mathbf{x}_\perp) = (g\rho(x^-, \mathbf{x}_\perp), \sim 0, \sim 0, \sim 0) \tag{2.15}$$

where the charge distribution is taken to be very narrowly peaked around $x^- = 0$.

2.5.3 The stochastic nature of ρ

The color charge distribution ρ, from which J^μ and, consequently, the evolution of the gluon fields are obtained, is fixed for a single nucleus in a single event. It is different, however, for different nuclei and for each event. For this reason, it is treated as a stochastic variable [7] which is neither accessible experimentally for each collision nor computable from first principles. Even considering a single nucleus, ρ would change if enough time is allowed to pass in order for the nucleus to travel over a x^+ length greater than the valence partons' typical x^+ scale [14]. In order to calculate an observable $\mathcal{O}$ out of many events within the CGC effective theory, one must determine its average value considering all the different possible color charge distributions and their probabilities: [7]

$$\langle \mathcal{O} \rangle = \int [\mathrm{d}\rho] W[\rho] \mathcal{O}[\rho] \tag{2.16}$$

where $\mathcal{O}[\rho]$ is the observable's value for the particular charge distribution ρ, and $W[\rho]$ is a functional probability distribution evaluated for that same charge distribution, describing its probability of occurring.

2.5.4 Quantum Corrections and the McLerran-Venugopalan Model

So far, the framework's dependence on the arbitrary choice of a $+$-momentum scale Λ^+ which separates low- and high-x partons has been neglected. Indeed, once a $+$-momentum scale has been set, one neglects that certain low-x partons sorted into the category of classical fields may actually act as static color sources for lower-x-still partons[6] [8]. Thus, choosing a different Λ^+ would lead to different final-state observable values, amounting to a quantum correction to the classical theory. Consider

[6] In the language of Section 2.2, even a low-x gluon can serve as an emitter of lower-x gluons.

changing the momentum scale from an initial value of Λ_+ to Λ'_+ – this alters the value of final-state observables calculated with the CGC framework. However, it is possible to renormalize ρ, correlators of ρ or the probability distribution W so as to incorporate these quantum corrections into the CGC effective theory. This preserves the value of the observables, making them independent of Λ^+ while keeping the calculation procedure strictly classical [8]. Choosing to incorporate the quantum corrections into W, which then becomes dependent on the $+$-momentum scale, W_{Λ^+}, its evolution with Λ^+ is ruled by the Jalilian-Marian–Iancu–McLerran–Weigert–Leonidov–Kovner (JIMWLK) equation (read "Jim-Walk") [7].

In spite of this achievement, an initial probability distribution $W_{\Lambda_0^+}$ is still necessary before studying its evolution with the $+$-momentum scale Λ^+; this function can be seen as an initial condition for the JIMWLK equation. The MV model, the founding iteration of the CGC, provides that initial condition, W_{MV}, for the case of large nuclei.

The MV model's functional probability distribution W_{MV} is motivated physically by the characteristics of large nuclei: [14]

1) Due to color confinement, color charge within different nucleons should be uncorrelated. Since charge densities at different transverse positions $\mathbf{x}_\perp$ and $\mathbf{y}_\perp$ are associated with different nucleons/longitudinal superpositions of nucleons, $\rho(\mathbf{x}_\perp)$ and $\rho(\mathbf{y}_\perp)$ should also be uncorrelated.

2) For large nuclei, at least at central positions, transverse color charge density results from the sum of many longitudinally aligned nucleons. Since the color charge of those nucleons is uncorrelated, then $\rho(\mathbf{x}_\perp)$ is a sum of uncorrelated variables. Therefore, the Central Limit Theorem applies, and $\rho(\mathbf{x}_\perp)$ should follow a Gaussian distribution.

The form of W_{MV}, apart from a normalization constant C, is [8] [15]:

$$W_{\text{MV}}[\rho] = C \exp\left[-\int \mathrm{d}x^- \, \mathrm{d}^2\mathbf{x}_\perp \frac{\rho^a(x^-,\mathbf{x}_\perp)\rho^a(x^-,\mathbf{x}_\perp)}{2\mu(x^-)^2}\right] \tag{2.17}$$

which has an associated charge density 2-point correlator:

$$\left\langle \rho_a(x^-,\mathbf{x}_\perp)\rho_b(y^-,\mathbf{y}_\perp)\right\rangle = \delta^{ab}\delta^2(\mathbf{x}_\perp - \mathbf{y}_\perp)\delta(x^- - y^-)\mu_A^2\lambda(x^-,\mathbf{x}_\perp) \tag{2.18}$$

[Explain μ/λ].

In spite of not incorporating quantum corrections, the MV model may be used directly, without needing to be evolved into x-scales much smaller than Λ_0^+, for the study of pA or AA collisions where the values of x are not small enough to require that evolution. [7]

The classical MV model will be used, and quantum corrections will be left out of the scope of this work. The application of this formalism to the two-nuclei system of a HIC will be introduced over the next sections.

2.6 Solving The Yang-Mills Equations

2.6.1 One nucleus

First, the case of a single nucleus will be considered. Nucleus 1 is chosen, moving in the positive x^1 direction. Solutions to the Yang-Mills equations for a single nucleus moving along the light-cone were first found analytically in [16] for the classical case $D_\mu F^{\mu\nu} = J^\nu$.

In the present work, the Yang-Mills equations in the case of a single nucleus are initially solved in the Lorenz, or covariant, gauge, where $\partial_\mu A^\mu = 0$, following the more recent solving of the full equations from [17, Section 2.2][7]. The solution is then brought to light-cone gauge, where $A^+ = 0$, an axial gauge condition, following the procedure from [15, Section 3A]. Bringing the final solution to light-cone gauge is necessary, as this is the gauge where the parton distribution operators have the physical interpretation of quantifying the parton populations inside a hadron. [18][19, Section 5.1.1][20, Section 2.4] Thus, this gauge is required in order to use our previous insights concerning the low-x gluon populations within protons. Additionally, use of the light-cone gauge allows reducing the problem to two degrees of freedom (A^2 and A^3). [19, Section 5.1.1].

First of all, the equation explicitly enforcing the covariant conservation of color charge is added to the set of equations to solve. It is analogous to Classical Electrodynamics' equation of conservation of electric charge, and is obtained from the Yang-Mills equations as:

$$[D_\mu, F^{\mu\nu}] = J^\nu \implies [D_\nu, J^\nu] = [D_\nu, [D_\mu, F^{\mu\nu}]] = 0 \tag{2.19}$$

via the Jacobi identity and the $F^{\mu\nu} = -ig[D_\mu, D_\nu]$ identity for the gluon field strength tensor (see Appendix A.1 for a detailed derivation).

The conservation equation and The Yang-Mills equations (in the Lorenz gauge) are then arranged into the system of equations:

$$\begin{cases} [D_\mu, J^\mu] = 0 \\ [D_\mu, F^{\mu\nu}] = J^\nu \end{cases} \Leftrightarrow \begin{cases} \partial_\mu J^\mu = -ig[A_\mu, J^\mu] \\ \partial_\mu \partial^\mu A^\nu = J^\nu - ig[A_\mu, F^{\mu\nu} + \partial^\mu A^\nu] \end{cases} \tag{2.20}$$

A solution may be built by solving this system iteratively, for the fields and current expanded in terms of powers of the color number charge density, ρ. The terms of order (n) are obtained from those of lower orders by:

$$\begin{cases} \partial_\mu J^\mu_{(n)} = -ig[A_\mu, J^\mu]_{(n)} \\ \partial_\mu \partial^\mu A^\nu_{(n)} = J^\nu_{(n)} - ig[A_\mu, F^{\mu\nu} + \partial^\mu A^\nu]_{(n)} \end{cases} \tag{2.21}$$

Inputting $J^\mu_{(0)} = A^\mu_{(0)} = 0$ (since, in the absence of color charges, there are neither any color currents nor fields generated from them) and $J^\mu_{(1)} = g\delta^\mu_+ \rho(x^-, \mathbf{x}_\perp)$ (Eq. (2.15)), the top equation for $n = 1$ is automatically satisfied since $\partial_+ J^+_{(1)} = 0$. The bottom equation may be solved by requiring that $A^\mu_{(1)}$, like

[7] Note that this paper's covariant derivative definition is $D_\mu = \partial_\mu - igA_\mu$. However, this does not affect the final result.

$J^\mu_{(1)}$, be independent of x^+, and that it vanishes in the far past [17]:

$$\begin{aligned} &\partial_\mu \partial^\mu A^\nu_{(1)} = J^\nu_{(1)} \;\rightarrow\; -\nabla^2_\perp A^\nu_{(1)} = J^\nu_{(1)} \\ &\rightarrow\; A^\nu_{(1)}(x^-, \mathbf{x}_\perp) = -g\delta^\nu_+ \int \mathrm{d}^2\mathbf{y}_\perp G(\mathbf{x}_\perp - \mathbf{y}_\perp)\rho(x^-, \mathbf{y}_\perp) \end{aligned} \tag{2.22}$$

where $G(\mathbf{x}_\perp - \mathbf{y}_\perp)$ is a Green's function for the 2-dimensional Laplacian, $\nabla^2_\perp = -\partial_i\partial^i$ with $i = 2, 3$.

Moving to the $n = 2$ system, the right side of the top equation is once again 0 since $A^\mu_{(1)}$ and $J^\mu_{(1)}$ only have a $+$-component. For the bottom equation, its commutator vanishes because $A^\mu_{(1)}$ only has a $+$-component and because that component is independent of x^+. The system for $n = 2$ is, then,

$$\begin{cases} \partial_\mu J^\mu_{(2)} = 0 \\ \partial_\mu \partial^\mu A^\nu_{(2)} = J^\nu_{(2)} \end{cases} \tag{2.23}$$

Once again requiring that the current and fields vanish in the far past , the solution is $J^\mu_{(2)} = 0$ for the top and $A^\mu_{(2)} = 0$ for the bottom. This makes all the greater-order terms also 0, making the final solution equal to the first order solution:

$$J^\mu(x^-, \mathbf{x}_\perp) = g\delta^\mu_+ \rho(x^-, \mathbf{x}_\perp) \tag{2.24a}$$

$$A^\nu(x^-, \mathbf{x}_\perp) = -g\delta^\nu_+ \int \mathrm{d}^2\mathbf{y}_\perp G(\mathbf{x}_\perp - \mathbf{y}_\perp)\rho(x^-, \mathbf{y}_\perp) \tag{2.24b}$$

Finally, in order to move to light-cone gauge, where $A^+_{\text{LC}} = 0$, the appropriate gauge transformation matrix U must be found:

$$A^\mu_{\text{LC}} = U^\dagger A^\mu U + \frac{i}{g} U^\dagger \partial^\mu U \tag{2.25}$$

It may be found by solving the equation:

$$A^+_{\text{LC}} = 0 \;\Leftrightarrow\; U^\dagger A^+ U + \frac{i}{g} U^\dagger \partial^+ U = 0 \;\Leftrightarrow\; A^+ U + \frac{i}{g}\partial_- U = 0 \;\Leftrightarrow\; \partial_- U = igA^+ U \tag{2.26}$$

which admits a solution of the type $U(x^-, \mathbf{x}_\perp)$ and may be solved iteratively in its integral form:

$$U(x^-, \mathbf{x}_\perp) = U(x^-_0, \mathbf{x}_\perp) + ig \int_{x^-_0}^{x^-} \mathrm{d}z A^+(z, \mathbf{x}_\perp) U(z, \mathbf{x}_\perp) \tag{2.27}$$

$$\begin{aligned}
U(x^-,\mathbf{x}_\perp) = \lim_{n\to\infty} \Bigg(\mathbb{1} \ &+ ig\int_{x^0_-}^{x^-} \mathrm{d}z_0 A^+(z_0) + (ig)^2 \int_{x^0_-}^{x^-} \mathrm{d}z_0 \int_{x^0_-}^{z_0} \mathrm{d}z_1 A^+(z_0)A^+(z_1) + ... \\
&+ (ig)^n \int_{x^0_-}^{x^-} \mathrm{d}z_0 \int_{x^0_-}^{z_0} \mathrm{d}z_1 ... \int_{x^0_-}^{z_{n-1}} \mathrm{d}z_n \ A^+(z_0)A^+(z_1)...A^+(z_n) \Bigg) U(x_0^-,\mathbf{x}_\perp) \\
= \lim_{n\to\infty} \Bigg(\mathbb{1} \ &+ ig\int_{x^0_-}^{x^-} \mathrm{d}z_0 A^+(z_0) + \frac{(ig)^2}{2} \int_{x^0_-}^{x^-} \mathrm{d}z_0 \int_{x^0_-}^{x^-} \mathrm{d}z_1 \mathcal{P}^-\Big\{ A^+(z_0)A^+(z_1) \Big\} + ... \\
&+ \frac{(ig)^n}{n!} \int_{x^0_-}^{x^-} \mathrm{d}z_0 \int_{x^0_-}^{x^-} \mathrm{d}z_1 ... \int_{x^0_-}^{x^-} \mathrm{d}z_n \ \mathcal{P}^-\Big\{ A^+(z_0)A^+(z_1)...A^+(z_n) \Big\} \Bigg) U(x_0^-,\mathbf{x}_\perp) \\
\equiv \mathcal{P}^- \exp &\left[ig\int_{x_0^-}^{x^-} \mathrm{d}z A^+(z,\mathbf{x}_\perp) \right] U(x_0^-,\mathbf{x}_\perp)
\end{aligned} \tag{2.28}$$

where the dependence of the A^+ fields on $\mathbf{x}_\perp$ was only shown explicitly in the last line and $\mathcal{P}^-$ is the path-ordering operator for the x^- variable; that is, it orders the A^+ field operators from right to left in descending order of their x^- argument:

$$x_n^- \geq x_{n-1}^- \geq ... \geq x_1 \implies \mathcal{P}^- \left\{ A^+(x^-_{p_n})A^+(x^-_{p_{n-1}})...A^+(x_{p_1}) \right\} \equiv A^+(x_n^-)A^+(x_{n-1}^-)...A^+(x_1) \tag{2.29}$$

with $\{p_n, p_{n-1}, ..., p_1\}$ some permutation of $\{n, n-1, ..., 1\}$

Additionally, $\mathcal{P}^- \exp[ig\int_{x_0^-}^{x^-} \mathrm{d}zA^+(z,\mathbf{x}_\perp)]$ is called the path-ordered exponential of the $ig\int_{x_0^-}^{x^-} \mathrm{d}zA^+(z,\mathbf{x}_\perp)$ integral.

This solution corresponds to a Wilson Line,

$$\mathcal{P}^- \exp\left[ig\int_{x_0^-}^{x^-} \mathrm{d}zA^+(z,\mathbf{x}_\perp) \right] \equiv W[A^+](x^-, x_0^-, \mathbf{x}_\perp) \tag{2.30}$$

These objects frequently appear in calculations in the CGC framework. One physical interpretation of a Wilson Line is as the matrix for the rotation in color space that a high-energy quark experiences when traversing a color field eikonally – that is, having negligible momentum exchange with the color field [21]. Indeed, one way to turn the initial expression for the color current (Eq. (2.15)) into a covariantly conserved one is to add Wilson line operators to it reflecting the color rotation it suffers as it traverses the gluon fields: $J^\mu(x^-,\mathbf{x}_\perp) \to W[A^-](x^+, x_0^+, \mathbf{x}_\perp)J^\mu(x^-,\mathbf{x}_\perp)W[A^-](x_0^+, x^+, \mathbf{x}_\perp)$. Here, the Wilson lines serve as link operators bringing the original (at x_0^+) current, J^μ, to the present, x^+.

It is possible to choose x_0^- to be $-\infty$ and $U(x_0^-,\mathbf{x}_\perp)$ to be $\mathbb{1}$ with U remaining a solution:

$$U(x^-,\mathbf{x}_\perp) = \mathcal{P}^- \exp\left[ig\int_{-\infty}^{x^-} \mathrm{d}zA^+(z,\mathbf{x}_\perp) \right] \tag{2.31}$$

Renaming the $+$-component of A in the Lorenz gauge to $\Phi(x^-,\mathbf{x}_\perp)$, the expressions for the gluon fields in light-cone gauge become:

$$A^+_{\mathsf{LC}} = 0 \tag{2.32a}$$

$$A^-_{\mathsf{LC}} = 0 \tag{2.32b}$$

$$A^i_{\mathsf{LC}} = \frac{i}{g} U(x^-, \mathbf{x}_\perp)^\dagger \, \partial^i U(x^-, \mathbf{x}_\perp), \qquad i = 1, 2 \tag{2.32c}$$

$$\begin{aligned} \text{with } U(x^-, \mathbf{x}_\perp) &= \mathcal{P}^- \exp\left[ig \int_{-\infty}^{x^-} \mathrm{d}z \; \Phi(z, \mathbf{x}_\perp) \right] \\ &= \mathcal{P}^- \exp\left[ig \int_{-\infty}^{x^-} \mathrm{d}z \; (-g) \int \mathrm{d}^2\mathbf{y}_\perp G(\mathbf{x}_\perp - \mathbf{y}_\perp) \rho(z, \mathbf{y}_\perp) \right] \end{aligned}$$

where the A^i fields, named the transverse fields, are pure-gauge fields due to them arising solely out of the gauge transformation term involving only U (Eq. (2.25)).

So far, all calculations have been performed keeping a generic, unknown dependence of ρ on x^-. However, remembering that the color charge density has very narrow support around $x^- = 0$:

$$\rho(x^-, \mathbf{x}_\perp) \approx \delta(x^-) \rho(\mathbf{x}_\perp) \tag{2.33}$$

the U operator's dependence on x^- turns out to be of the form:

$$U(x^-, \mathbf{x}_\perp) = \begin{cases} \mathbb{1}, & x^- < 0 \\ U(+\infty, \mathbf{x}_\perp), & x^- > 0 \end{cases} \tag{2.34}$$

which allows factoring the x^- dependence of the A^i_{LC} fields:

$$A^i_{\mathsf{LC}}(x^-, \mathbf{x}_\perp) \equiv \theta(x^-) \alpha^i_{\mathsf{LC}}(\mathbf{x}_\perp) \tag{2.35}$$

This factorization is possible due to the choice $U(-\infty, \mathbf{x}_\perp) = \mathbb{1}$. Another, $\mathbf{x}_\perp$-dependent form for $U(-\infty, \mathbf{x}_\perp)$ may be chosen, amounting to a further gauge transformation $U'(\mathbf{x}_\perp)$ which keeps A'^+ and A'^- equal to 0 (and consequently keeps the solution in the light-cone gauge), but due to which the transverse fields A'^i do not vanish for $x^- < 0$. However, keeping the fields in the form of Eq. (2.35) will be useful for establishing boundary conditions for the resolution of the Yang-Mills equations in the space between the two nuclei, after the collision. Addditionally, it corresponds to having the gluon fields vanish in the region in front of the planar nucleus; since the nucleus travels at the speed of light, the region in front of it should not "feel" the presence of its color charges due to causality, and this is explicitly the case in the present gauge. $\times$

One last topic that should be discussed before moving on to the situation of the oppositely-moving nucleus is that, unlike in the abelian case of Classical Electrodynamics, in the present case of a classical non-abelian gauge theory of Chromodynamics gauge transformations alter the distributions of color charge; otherwise, the introduced Lagrangian, Eq. (2.11), would not be invariant under SU(3) gauge transformations. Thus, one should remember that the color charge distribution ρ discussed in this chapter and present in its final result (Eqs. (2.32)) is the color charge distribution in the Lorenz, and not the light-cone, gauge. In the rest of this work, the subscript "LC" denoting results in the light-cone gauge will be dropped; in order to avoid confusion, color number charge densities in the Lorenz gauge will be denoted by $\tilde{\rho}$ instead of ρ.

The previous discussion of the fields generated by a single nucleus traveling at the speed of light in the positive x^1 direction (Nucleus 1) can be easily adapted for a single nucleus traveling in the negative x^1 direction (Nucleus 2). Labeling all quantities in this scenario with the subscript 2, equivalent arguments to those used in Section 2.5.2 lead to a color current with only a $-$-component, of the form:

$$J_2^\mu = g\delta_-^\mu \tilde{\rho}_2(x^+, \mathbf{x}_\perp) \approx g\delta_-^\mu \delta(x^+)\tilde{\rho}_2(\mathbf{x}_\perp) \tag{2.36}$$

Which generates the gluon fields, in the light-cone gauge[8]

$$A_2^+ = 0 \tag{2.37a}$$

$$A_2^- = 0 \tag{2.37b}$$

$$A_2^i = \frac{i}{g} U_2(x^+, \mathbf{x}_\perp)^\dagger \, \partial^i U_2(x^+, \mathbf{x}_\perp) \equiv \theta(x^+)\alpha_2^i(\mathbf{x}_\perp), \qquad i = 1, 2 \tag{2.37c}$$

with

$$U_2(x^+, \mathbf{x}_\perp) = \mathcal{P}^+ \exp\left[ig \int_{-\infty}^{x^+} \mathrm{d}z \; \Phi_2(z, \mathbf{x}_\perp)\right] = \mathcal{P}^+ \exp\left[ig \int_{-\infty}^{x^+} \mathrm{d}z \; (-g) \int \mathrm{d}^2\mathbf{y}_\perp G(\mathbf{x}_\perp - \mathbf{y}_\perp)\tilde{\rho}_2(z, \mathbf{y}_\perp)\right]$$

Expressions for the gluon fields around a single nucleus traveling at the speed of light have finally been obtained. Next, when studying the HIC system of two colliding nuclei, they will be useful for evaluating the gluon fields in regions of spacetime causally related to only one of the nuclei, and for setting the boundary conditions for solving the Yang-Mills equations everywhere else.

2.6.2 Two nuclei

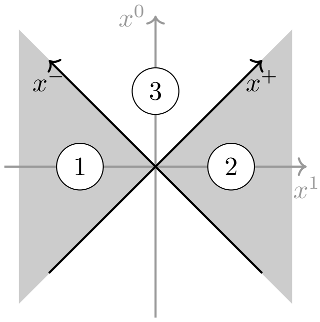
Moving to the scenario of a HIC, where both nuclei are present, the recent solvings in [22, 23] will be followed. The use of a CGC formalism to study the case of a HIC was first accomplished in [18].

The two currents are added, and J^μ is given by the expression:

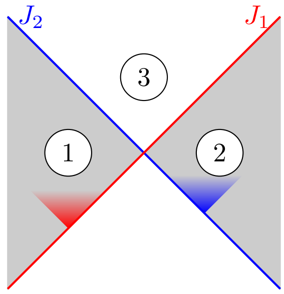
$$\begin{aligned} J^\mu(x^\mu) &= J_1^\mu(x^-, \mathbf{x}_\perp) + J_2^\mu(x^+, \mathbf{x}_\perp) = g\delta_+^\mu \tilde{\rho}_1(x^-, \mathbf{x}_\perp) + g\delta_-^\mu \tilde{\rho}_2(x^+, \mathbf{x}_\perp) \\ &\approx g\delta_+^\mu \delta(x^-)\tilde{\rho}_1(\mathbf{x}_\perp) + g\delta_-^\mu \delta(x^+)\tilde{\rho}_2(\mathbf{x}_\perp) \end{aligned} \tag{2.38}$$

For the purpose of solving the Yang-Mills equations, spacetime is divided into four quadrants in (x^+, x^-) space, as illustrated in Fig. 2.7a. Quadrants 1 and 2 are named the pre-collision quadrants, while quadrant 3 is referred to as the forward light-cone. For calculations in the third quadrant, the Milne

[8] In this case, the light-cone gauge condition is $A^- = 0$.



(a) Partition of spacetime into light-cone quadrants. The quadrants where the Yang-Mills equations will be solved are identified as quadrants 1, 2 and 3. Adapted from [18].



(b) The same four quadrants with the support of current J_1 (J_2) drawn as a red (blue) line. Future light-cones drawn from points where J_1 passes only occupy quadrants 1 and 3; future light-cones drawn from points where J_2 passes only occupy quadrants 2 and 3.

Figure 2.7

coordinate system[9] is used, where positions and four-vectors are given by:

$$\tau = \sqrt{2x^+x^-} \tag{2.39a}$$

$$\eta = \frac{1}{2}\log\left(\frac{x^+}{x^-}\right) \tag{2.39b}$$

$$A^\tau = \frac{x^-A^+ + x^+A^-}{\tau} \tag{2.40a}$$

$$A^\eta = \frac{x^-A^+ - x^+A^-}{\tau^2} \tag{2.40b}$$

and the transverse components ($i = 2, 3$) keep their Cartesian form. τ is called the proper time and η is called the rapidity. For a more detailed overview of the Comoving coordinates, see Appendix B.2.

Drawing future light-cones out from points where the color currents pass in (x^+, x^-) space, one notices that all points in quadrant 1 may not be affected by the J_2 current, due to causality; in the same manner, all points in quadrant 2 may not be affected by current J_1. Only points in quadrant 3 are causally related to both currents (Fig. 2.7b). Following [22]'s approach, this permits the "single-nuclei" results of the previous section to be used for the gluon fields in quadrants 1 and 2, reducing the problem to solving the Yang-Mills equations in the third quadrant. Even though they were obtained using different gauge conditions ($A^+ = 0$ for the first nucleus and $A^- = 0$ for the second nucleus), the result in Eqs. (2.32) and (2.35) can be used for the fields in quadrant 1 and the result in Eqs. (2.37) can be used for the fields in quadrant 2, since those regions are not causally related. In solving the Yang Mills equations for the case of two nuclei, the Fock-Schwinger gauge is chosen, defined by the axial gauge condition:

$$x^+A^- + x^-A^+ = 0 \;\Leftrightarrow\; A^\tau = 0 \tag{2.41}$$

which is obeyed by the solutions built for quadrants 1 and 2. Moving on to quadrant 3, in its interior, this

[9] Also called the Bjorken or Comoving coordinate system.

gauge choice requires A^+ and A^- to be related as:

$$A^- = -\frac{x^-}{x^+}A^+ \tag{2.42}$$

which suggests the definition of an α field as:

$$A^+ \equiv x^+\alpha \implies A^- = -x^-\alpha \tag{2.43}$$

Inserting these definitions in Eq. (2.40b), the α field can be shown to be equal to A^η.

Following the ansatz first proposed in [18], the α field and the transverse fields of the third quadrant, α^i, are taken not to depend on η. This is so that the form of the solution in the third quadrant is not altered by applying a boost to the studied system along the direction of motion of the two nuclei, since the system is invariant under boosts along the x^1 axis. For any boost applied, the two infinitely thin nuclei will keep moving at the speed of light, making the boosted system indistinguishable from the original and requiring a solution of the same form. Compiling all the information gathered so far regarding the form of the solution,

$$A^+(x^\mu) = \underbrace{\theta(x^+)\theta(x^-)}_{③}\,x^+\alpha(\tau,\mathbf{x}_\perp) \tag{2.44a}$$

$$A^-(x^\mu) = -\underbrace{\theta(x^+)\theta(x^-)}_{③}\,x^-\alpha(\tau,\mathbf{x}_\perp) \tag{2.44b}$$

$$A^i(x^\mu) = \underbrace{\theta(-x^+)\theta(x^-)}_{①}\,\alpha_1^i(\mathbf{x}_\perp) + \underbrace{\theta(x^+)\theta(-x^-)}_{②}\,\alpha_2^i(\mathbf{x}_\perp) + \underbrace{\theta(x^+)\theta(x^-)}_{③}\,\alpha^i(\tau,\mathbf{x}_\perp) \tag{2.44c}$$

For the third quadrant (terms underbraced with '3'), applying such a boost as described leaves the form of the solution unchanged, as τ and $\mathbf{x}_\perp$ are invariant under it.

In the interior of the third quadrant, the currents J_1^μ and J_2^μ have no support, and the Yang-Mills equations take the form:

$$[D_\mu, F^{\mu\nu}] = 0 \tag{2.45}$$

Before moving on to their treatment, a discussion of the boundary conditions to be used to connect the solution in the forward light-cone with the already-found solutions in the pre-collision quadrants is in order. It is possible to obtain, out of the already-found fields in the pre-collision quadrants, starting values for the fields in the forward light-cone at $\tau = 0^+$. This is usually done by requiring that, using the ansatz in Eq. (2.44), the Yang-Mills equations be regular at $\tau = 0$ [23]. They may also be obtained by first considering that the longitudinal width of the sources is finite and greater than 0, rather than infinitely thin as in Eq. (2.33), then integrating the Yang-Mills equations over a path (Eq. (2.46a)) and an area (Eq. (2.46b)) which cross the light-cone, and finally taking the limit where the width of the sources goes to zero. [24, Appendix C] The obtained boundary conditions are:

$$\alpha(\tau \to 0^+, \mathbf{x}_\perp) = \frac{ig}{2}\left[\alpha_1^i(\mathbf{x}_\perp), \alpha_2^i(\mathbf{x}_\perp)\right] \tag{2.46a}$$

$$\alpha^i(\tau \to 0^+, \mathbf{x}_\perp) = \alpha_1^i(\mathbf{x}_\perp) + \alpha_2^i(\mathbf{x}_\perp) \tag{2.46b}$$

Another way to express these boundary conditions is via the chromodynamic force fields: the electric chromodynamic field, $\mathbf{E}$, and the magnetic chromodynamic field, $\mathbf{B}$, which will be referred to simply as the electric and magnetic fields. In Minkowski coordinates, the components of the chromodynamic fields are obtained from the components of the gluon field strength tensor, $F^{\mu\nu}$, following [25], as:

$$E^i = F_{0i} \tag{2.47a}$$

$$B^i = -\frac{1}{2}\varepsilon^{ijk}F_{jk} \tag{2.47b}$$

with $i, j, k = 1, 2, 3$, as in the case of Classical Electrodynamics. In Milne coordinates, the components are defined as:

$$E^\eta = \frac{1}{\tau}F_{\tau\eta} = -F_{+-} \qquad E^i = F_{\tau i} \tag{2.48a}$$

$$B^\eta = -\frac{1}{2}\varepsilon^{ij}F_{ij} = -F_{23} \quad B^i = -\varepsilon^{ij}\frac{1}{\tau}F_{j\eta} \tag{2.48b}$$

with $i, j = 2, 3$. Defined this way, $E^\eta = E^1$ and $B^\eta = B^1$ everywhere, earning them the name "longitudinal fields"; on the other hand, $E^i_{\text{Minkowski}} = E^i_{\text{Milne}}$ and $B^i_{\text{Minkowski}} = B^i_{\text{Milne}}$ ($i = 2, 3$) at midrapidity – these are the transverse fields.

The boundary conditions for the electric and magnetic fields are obtainable by calculating the relevant components of $F^{\mu\nu}$ using the boundary conditions already found for α and α^i (Eqs. (2.46)):

$$E_0^\eta(\mathbf{x}_\perp) = -ig\delta^{ij}\left[\alpha_1^i(\mathbf{x}_\perp), \alpha_2^j(\mathbf{x}_\perp)\right] \tag{2.49a}$$

$$B_0^\eta(\mathbf{x}_\perp) = -ig\varepsilon^{ij}\left[\alpha_1^i(\mathbf{x}_\perp), \alpha_2^j(\mathbf{x}_\perp)\right] \tag{2.49b}$$

Additionally, the initial chromodynamic fields are purely longitudinal; that is, the transverse fields are initially null, $E_0^i = B_0^i = 0$. The steps of this derivation are detailed in Appendix A.3.

Moving on to the treatment of the Yang-Mills equations, defining $Y^\nu \equiv [D_\mu, F^{\mu\nu}]$, the τ and η components of the system of equations in comoving coordinates may be obtained out of the $+$ and $-$ components in light-cone coordinates as:

$$Y^\tau = 0 \;\Leftrightarrow\; \frac{x^- Y^+ + x^+ Y^-}{\tau} = 0 \tag{2.50a}$$

$$Y^\eta = 0 \;\Leftrightarrow\; \frac{x^- Y^+ - x^+ Y^-}{\tau^2} = 0 \tag{2.50b}$$

and the $\nu = 2, 3$ components keep their cartesian form.

Inserting Eq. (2.44) into the Yang-Mills equations in comoving coordinates returns:

$$Y^\tau = 0 \quad \Leftrightarrow \quad ig\tau\left[\alpha, \partial_\tau \alpha\right] - \frac{1}{\tau}\left[D_i, \partial_\tau \alpha^i\right] = 0 \tag{2.51a}$$

$$Y^\eta = 0 \quad \Leftrightarrow \quad \frac{1}{\tau}\partial_\tau\left(\frac{1}{\tau}\partial_\tau\left(\tau^2\alpha\right)\right) - \left[D^i, \left[D^i, \alpha\right]\right] = 0 \tag{2.51b}$$

$$Y^i = 0 \quad \Leftrightarrow \quad \frac{1}{\tau}\partial_\tau\left(\tau\partial_\tau\alpha^i\right) - ig\tau^2\left[\alpha, \left[D^i, \alpha\right]\right] - \left[D^j, F^{ji}\right] = 0 \tag{2.51c}$$

where $i, j = 2, 3$.

These equations don't look very friendly. They have no known analytical solution, but have been studied numerically (say more) and analytically via approximations (say more). In this work, the approximation that will allow the analytical treatment of the equations is their linearization; that is, the neglect of second or higher order terms in the fields, which was first done in []. This will naturally do away with gauge invariance, as the value of the neglected terms changes according to the used gauge. Therefore, one must choose the proper gauge, which minimizes the influence of these terms, before linearizing the equations. Two criteria are put forward in [23]: keeping the result independent of η suggests that the gauge transformation matrix should be independent of η as well; additionally, since there are no color charge sources in the interior of the forward light-cone that would generate a static Coulomb field, the Fock-Schwinger gauge condition ($A^\tau = 0$) is maintained[10], which can be achieved by requiring the gauge transformation to be independent of τ. Thus, the inverse gauge transformation is:

$$\alpha(\tau, \mathbf{x}_\perp) = V(\mathbf{x}_\perp)\beta(\tau, \mathbf{x}_\perp)V^\dagger(\mathbf{x}_\perp) \tag{2.52a}$$

$$\alpha^i(\tau, \mathbf{x}_\perp) = V(\mathbf{x}_\perp)\left(\beta(\tau, \mathbf{x}_\perp) - \frac{1}{ig}\partial^i\right)V^\dagger(\mathbf{x}_\perp) \tag{2.52b}$$

which makes the linearized Yang-Mills equations for the fields in the new gauge, CHECK

$$\partial_\tau\partial_i\beta^i = 0 \tag{2.53a}$$

$$\frac{1}{\tau}\partial_\tau\left(\frac{1}{\tau}\partial_\tau\left(\tau^2\beta\right)\right) - \partial^i\partial^i\beta = 0 \tag{2.53b}$$

$$\frac{1}{\tau}\partial_\tau\left(\tau\partial_\tau\beta^i\right) - \left(\partial^k\partial^k\delta^{ij} - \partial^j\partial^i\right)\beta^j = 0 \tag{2.53c}$$

where $k = 2, 3$. (different signs in 1808 and 2102)

The remaining gauge freedom allows setting $\partial_i\beta^i = 0$, a Coulomb gauge condition, minimizing the amplitudes of the transverse fields [26]. The equations become:

$$\partial_\tau\left(\frac{1}{\tau}\partial_\tau\left(\tau^2\beta\right)\right) - \tau\partial^i\partial^i\beta = 0 \tag{2.54a}$$

$$\partial_\tau\left(\tau\partial_\tau\beta^i\right) - \tau\partial^k\partial^k\beta^i = 0 \tag{2.54b}$$

[10] Maintaining the Fock-Schwinger gauge also ensures that A^- remains 0 in the support of the right-moving current and A^+ remains 0 in the support of the left-moving current, doing away with the complicated Wilson Line operators that would be needed to ensure their covariant conservation.

Taking the transverse position Fourier transform on both sides,

$$\beta(\tau,\mathbf{x}_\perp) = \iint \frac{\mathrm{d}^2\mathbf{k}_\perp}{(2\pi)^2}\, e^{i(k_\perp)^j(x_\perp)_j}\tilde{\beta}(\tau,\mathbf{k}_\perp) \tag{2.55a}$$

$$\beta^i(\tau,\mathbf{x}_\perp) = \iint \frac{\mathrm{d}^2\mathbf{k}_\perp}{(2\pi)^2}\, e^{i(k_\perp)^j(x_\perp)_j}\tilde{\beta}^i(\tau,\mathbf{k}_\perp) \tag{2.55b}$$

the equations for $\tilde{\beta}$ and $\tilde{\beta}^i$ are:

$$\tau^2\partial_\tau^2\big(\tau\tilde{\beta}\big) + \tau\partial_\tau\big(\tau\tilde{\beta}\big) + \big(\tau^2k^2-1\big)\big(\tau\tilde{\beta}\big) = 0 \tag{2.56a}$$

$$\tau^2\partial_\tau^2\tilde{\beta}^i + \tau\partial_\tau\tilde{\beta}^i + \tau^2k^2\tilde{\beta}^i = 0 \tag{2.56b}$$

where $k \equiv |\mathbf{k}_\perp| = \sqrt{(k^2)^2+(k^3)^2}$ is the Euclidean norm of $\mathbf{k}_\perp$.

Once the change of variable $z = k\tau$ is made, Eq. (2.56a) is a Bessel differential equation of degree 1 for the function $\tau\tilde{\beta}$. It admits as solutions linear combinations of the Bessel functions of order 1 of the first and second kind, $J_1(k\tau)$ and $Y_1(k\tau)$. Similarly, after the same change of variable, Eq. (2.56b) is a Bessel differential equation of degree 0 for the function $\tilde{\beta}^i$, which admits as solutions linear combinations of the Bessel functions of order 0 of the first and second kind, $J_0(k\tau)$ and $Y_0(k\tau)$. [27] As the Bessel functions of the second kind, Y_n, are singular at the origin for integer n, they are taken as unphysical solutions and their coefficients are set to 0.

The solution is, then,

$$\tilde{\beta}(\tau,\mathbf{k}_\perp) = \tilde{\beta}_0(\mathbf{k}_\perp)\frac{2J_1(k\tau)}{k\tau} \tag{2.57a}$$

$$\tilde{\beta}^i(\tau,\mathbf{k}_\perp) = \tilde{\beta}^i_0(\mathbf{k}_\perp)J_0(k\tau) \tag{2.57b}$$

where the factor of 2 in Eq. (2.57a) was introduced in order to make the $\mathbf{k}_\perp$-dependent coefficient, $\tilde{\beta}_0$, match the value of $\tilde{\beta}$ at $\tau = 0$, since $\lim_{z\to 0} J_1(z)/z = 1/2$. [27]

In order to find the expressions for the fields at $\tau = 0$ ($\tilde{\beta}_0$ and $\tilde{\beta}^i_0$), the next step would be to use the boundary conditions of Eqs. (2.46), which would require first bringing the pre-collision potentials, α^i_1 and α^i_2, to the Coulomb gauge. However, no analytical expression for the necessary transformation matrix $V(\mathbf{x}_\perp)$ has been found. In spite of this, another approach to this problem has found that the resolution of the linearized equations with numerically-computed boundary conditions in the Coulomb gauge approximate the resolution of the full Yang-Mills equations quite well [23, 26] , except for very soft modes (very low $|\mathbf{k}_\perp|$).

One way to go around this problem is to recall that the quantities needed to compute the physically meaningful results are the chromodynamic force fields: the electric field, $\mathbf{E}$, and the magnetic field, $\mathbf{B}$. They will be used to calculate color traces of products of their components, like $\mathrm{Tr}_c[E^i(\tau,\mathbf{x}_\perp)B^j(\tau',\mathbf{y}_\perp)]$, for example. Checking for gauge invariance, since the field components change under the latest gauge transformation as $E^i(\tau,\mathbf{x}_\perp) \to E'^i(\tau,\mathbf{x}_\perp) = V(\mathbf{x}_\perp)E^i(\tau,\mathbf{x}_\perp)V^\dagger(\mathbf{x}_\perp)$ (NOT GAUGE INVARIANT FOR $v_\perp \neq 0$?? , the previous trace is gauge invariant.

As such, in order to obtain the initial fields $\tilde{\beta}_0$ and $\tilde{\beta}^i_0$, one may choose them so as to produce the appropriate initial chromodynamic fields given by Eqs. (2.49): $\mathbf{E}_0(\mathbf{x}_\perp) \equiv \mathbf{E}(\tau \to 0^+, \mathbf{x}_\perp)$ and $\mathbf{B}_0(\mathbf{x}_\perp) \equiv \mathbf{B}(\tau \to 0^+, \mathbf{x}_\perp)$.

The relations between the initial gluon fields and the initial chromodynamic fields are obtained from taking the Fourier transform of the gluon and chromodynamic fields in Eqs. (2.48), as detailed in Appendix A.2:

$$\tilde{E}_0^\eta(\mathbf{k}_\perp) = -2\tilde{\beta}_0(\mathbf{k}_\perp) \implies \tilde{\beta}_0(\mathbf{k}_\perp) = -\frac{1}{2}\tilde{E}_0^\eta(\mathbf{k}_\perp) \tag{2.58a}$$

$$\tilde{B}_0^\eta(\mathbf{k}_\perp) = -i\varepsilon^{ij}k^i\tilde{\beta}_0^j(\mathbf{k}_\perp) \implies \tilde{\beta}_0^i(\mathbf{k}_\perp) = -i\varepsilon^{ij}\frac{k^j}{|\mathbf{k}_\perp|^2}\tilde{B}_0^\eta(\mathbf{k}_\perp) \tag{2.58b}$$

Thus, the chain of expressions which allows writing the chromodynamic fields as functions of the pre-collision potentials is now complete.

$$\tilde{E}^\eta(\tau, \mathbf{k}_\perp) = \tilde{E}_0^\eta(\mathbf{k}_\perp)J_0(k\tau) \tag{2.59a}$$

$$\tilde{E}^i(\tau, \mathbf{k}_\perp) = -i\varepsilon^{ij}(k^j/k)\tilde{B}_0^\eta(\mathbf{k}_\perp)J_1(k\tau) \tag{2.59b}$$

$$\tilde{B}^\eta(\tau, \mathbf{k}_\perp) = \tilde{B}_0^\eta(\mathbf{k}_\perp)J_0(k\tau) \tag{2.59c}$$

$$\tilde{B}^i(\tau, \mathbf{k}_\perp) = -i\varepsilon^{ij}(k^j/k)\tilde{E}_0^\eta(\mathbf{k}_\perp)J_1(k\tau) \tag{2.59d}$$

In order to obtain these expressions in position space, it is necessary to calculate the Fourier transforms of Eqs. (2.49) to insert into Eqs. (2.59), before applying the Fourier transform again to finish at position space. For example, for E^i:

$$\begin{aligned} E^i(\tau, \mathbf{x}_\perp) &= \int \frac{\mathrm{d}^2\mathbf{k}_\perp}{(2\pi^2)} e^{i\mathbf{k}_\perp\cdot\mathbf{x}_\perp}(-i)\varepsilon^{ij}\frac{k^j}{k}J_1(k\tau)\int \mathrm{d}^2\mathbf{u}_\perp e^{-i\mathbf{k}_\perp\cdot\mathbf{u}_\perp}(-ig)\varepsilon^{mn}[\alpha_1^m(\mathbf{u}_\perp)\alpha_2^n(\mathbf{u}_\perp)] \\ &= -ig\varepsilon^{mn}f^{abc}T_c\varepsilon^{ij}\int \frac{\mathrm{d}^2\mathbf{k}_\perp}{(2\pi^2)}\frac{k^j}{k}J_1(k\tau)\int \mathrm{d}^2\mathbf{u}_\perp e^{i\mathbf{k}_\perp\cdot(\mathbf{x}-\mathbf{u})_\perp}\alpha_{1a}^m(\mathbf{u}_\perp)\alpha_{2b}^n(\mathbf{u}_\perp) \end{aligned} \tag{2.60}$$

where the definition $\alpha_i^m(\mathbf{u}_\perp) \equiv \alpha_{ia}^m(\mathbf{u}_\perp)T_a$ and the SU(3) commutation relations, $[T_a, T_b] = if^{abc}T_c$, were used.

2.7 Transcribed expressions yet without text

In order to get expressions for one of the fields in coordinate space[11], the appropriate field at $\tau = 0^+$ (eqs. **??**) must be Fourier-transformed and used in Eqs. (2.59). The inverse Fourier transform is then applied to the resulting expression. For E^α, $\alpha = 2, 3$, for example:

[11] as is necessary in order to calculate $X^{\alpha\beta}$

$$
\begin{aligned}
E^{\alpha}(\tau,\mathbf{x}_{\perp}) &= \int \frac{\mathrm{d}^2\mathbf{k}_{\perp}}{(2\pi^2)} e^{i\mathbf{k}_{\perp}\cdot\mathbf{x}_{\perp}}\,(-i)\varepsilon^{\alpha\beta}\hat{k}^{\beta}J_1(k\tau)\int \mathrm{d}^2\mathbf{u}_{\perp}e^{-i\mathbf{k}_{\perp}\cdot\mathbf{u}_{\perp}} \\
&\qquad\qquad\qquad\qquad \times\ g\varepsilon^{ij}\alpha^{i}_{1a}(\mathbf{u}_{\perp})\alpha^{j}_{2b}(\mathbf{u}_{\perp})f^{abc}T_c \\
&= -ig\varepsilon^{ij}f^{abc}T_c\varepsilon^{\alpha\beta}\int \frac{\mathrm{d}^2\mathbf{k}_{\perp}}{(2\pi^2)}\hat{k}^{\beta}J_1(k\tau)\int \mathrm{d}^2\mathbf{u}_{\perp}e^{i\mathbf{k}_{\perp}\cdot(\mathbf{x}-\mathbf{u})_{\perp}} \\
&\qquad\qquad\qquad\qquad \times\alpha^{i}_{1a}(\mathbf{u}_{\perp})\alpha^{j}_{2b}(\mathbf{u}_{\perp})
\end{aligned}
\tag{2.61}
$$

And the remaining fields:

$$
E^{\eta}(\tau,\mathbf{x}_{\perp}) = g\delta^{ij}f^{abc}T_c\int \frac{\mathrm{d}^2\mathbf{k}_{\perp}}{(2\pi^2)}J_0(k\tau)\int \mathrm{d}^2\mathbf{u}_{\perp}e^{i\mathbf{k}_{\perp}\cdot(\mathbf{x}-\mathbf{u})_{\perp}}\alpha^{i}_{1a}(\mathbf{u}_{\perp})\alpha^{j}_{2b}(\mathbf{u}_{\perp}) \tag{2.62}
$$

$$
B^{\eta}(\tau,\mathbf{x}_{\perp}) = g\varepsilon^{ij}f^{abc}T_c\int \frac{\mathrm{d}^2\mathbf{k}_{\perp}}{(2\pi^2)}J_0(k\tau)\int \mathrm{d}^2\mathbf{u}_{\perp}e^{i\mathbf{k}_{\perp}\cdot(\mathbf{x}-\mathbf{u})_{\perp}}\alpha^{i}_{1a}(\mathbf{u}_{\perp})\alpha^{j}_{2b}(\mathbf{u}_{\perp}) \tag{2.63}
$$

$$
B^{\alpha}(\tau,\mathbf{x}_{\perp}) = -ig\delta^{ij}f^{abc}T_c\varepsilon^{\alpha\beta}\int \frac{\mathrm{d}^2\mathbf{k}_{\perp}}{(2\pi^2)}\hat{k}^{\beta}J_1(k\tau)\int \mathrm{d}^2\mathbf{u}_{\perp}e^{i\mathbf{k}_{\perp}\cdot(\mathbf{x}-\mathbf{u})_{\perp}}\alpha^{i}_{1a}(\mathbf{u}_{\perp})\alpha^{j}_{2b}(\mathbf{u}_{\perp}) \tag{2.64}
$$

Chapter 3

[Margaret]

[Likely to Migrate, requires references (textbooks?)]

The objectives of this work are: the derivation, by analytical means, of expressions for parameters characterizing the pre-equilibrium stage of a HIC as a medium which affects the highly energetic partons that traverse it; and the numerical evaluation of those expressions and comparison of the results obtained to others in the literature.

Those parameters are: the *stopping power*, denoted by $-\mathrm{d}E/\,\mathrm{d}x$, and the *momentum broadening coefficient*, denoted by $\hat{q}$.

In Particle Physics, the stopping power of a medium as felt by a probe traversing it at a given instant corresponds to the amount of kinetic energy being lost by the probe per unit length traveled in the medium. Less formally, it is equal to the infinitesimal loss of energy of the probe by interaction with the medium divided by the infinitesimal distance traversed during that loss:

$$-\frac{\mathrm{d}E}{\mathrm{d}x} \approx -\frac{\Delta E}{\Delta x}$$

The stopping power may be viewed as a measure of how capable a medium is of slowing down or stopping traversing particles within a given length, which makes it a quantity commonly discussed in the field of radiation protection, for example.

The momentum broadening coefficient, on the other hand, measures how capable a medium is of deflecting traversing particles, that is, of changing their original direction. It corresponds to the square magnitude of transverse momentum gained by the probe per unit length traveled in the medium, transverse meaning perpendicular to the probe's velocity in that instant.

$$\hat{q} \approx \frac{(\Delta \mathbf{p}_T)^2}{\Delta x}$$

These quantities are dependent on the properties of the medium and of the probe, on the velocity of the probe, and, in the case of a heterogeneous, dynamic medium, on the current position of the probe and on time.

The chromodynamic field correlators derived in the previous section may be used to obtain the momentum broadening coefficient, $\hat{q}$, and the stopping power, $\mathrm{d}E/\,\mathrm{d}x$, felt by a relativistic quark probe traversing the glasma. This procedure is outlined in [15].

3.1 The Fokker-Planck equation and its collision terms

The Fokker-Planck equation has frequently been used to study the transport of heavy quarks across a thermalized QGP. The following Fokker-Planck equation, derived in [28] for the transport of heavy quarks through a glasma, is also applicable to relativistic light partons, as long as the diffusion approximation is applicable [15], that is, as long as the transfer of momentum to the probe is negligible when compared to the probe's momentum [28]. It commands the evolution of the event-averaged probe quark distribution function $n(t, \mathbf{x}, \mathbf{p})$ as the probes traverse the medium.

$$\left(\frac{\partial}{\partial t} + \mathbf{v}\cdot\nabla - \nabla_p^\alpha X^{\alpha\beta}\nabla_p^\beta - \nabla_p^\alpha Y^\alpha\right) n(t, \mathbf{x}, \mathbf{p}) = 0 \tag{3.1}$$

where t is time, v is the probe's velocity, $\nabla^\alpha = (\partial_1, \partial_2, \partial_3)$ is the usual gradient, ∇_p^α is the gradient with respect to the probes' momentum, $\partial/(\partial p^\alpha)$, $\mathbf{x}$ and $\mathbf{p}$ are the probes' position and momentum, respectively, and α, β run from 1 through 3. The tensors $X^{\alpha\beta}$ and Y^α, from which $\hat{q}$ and $\mathrm{d}E/\,\mathrm{d}x$ will be calculated, are called the collision terms.

The collision terms are related to event-averaged[1] rates of change of the momentum of the probes over time:

$$\frac{\left\langle \Delta p^\alpha \Delta p^\beta \right\rangle}{\Delta t} = X^{\alpha\beta} + X^{\beta\alpha} \tag{3.2a}$$

$$\frac{\left\langle \Delta p^\alpha \right\rangle}{\Delta t} = -Y^\alpha \tag{3.2b}$$

As such, $\hat{q}$ and $\mathrm{d}E/\,\mathrm{d}x$ may be obtained from $X^{\alpha\beta}$ and Y^α:

$$\hat{q} = \frac{1}{v}\left(\delta^{\alpha\beta} - \frac{v^\alpha v^\beta}{v^2}\right)\frac{\left\langle \Delta p^\alpha \Delta p^\beta \right\rangle}{\Delta t} = \frac{2}{v}\left(\delta^{\alpha\beta} - \frac{v^\alpha v^\beta}{v^2}\right) X^{\alpha\beta} \tag{3.3a}$$

$$\frac{\mathrm{d}E}{\mathrm{d}x} = \frac{v^\alpha}{v}\frac{\left\langle \Delta p^\alpha \right\rangle}{\Delta t} = -\frac{v^\alpha}{v} Y^\alpha \tag{3.3b}$$

Should these auxiliary calcs be moved to an appendix? To prove the first equality of eq. 3.3a,

$$\frac{1}{v}\left(\delta^{\alpha\beta} - \frac{v^\alpha v^\beta}{v^2}\right)\frac{\left\langle \Delta p^\alpha \Delta p^\beta \right\rangle}{\Delta t} = \frac{1}{\Delta x}\left\langle (\Delta\mathbf{p})^2 - \left(\frac{\mathbf{v}\cdot\Delta\mathbf{p}}{v^2}\right)\right\rangle = \frac{\left\langle (\Delta\mathbf{p}_T)^2 \right\rangle}{\Delta x}$$

To prove the second equality of eq. 3.3a, eq. 3.2a is plugged in and the symmetry of $(\delta^{\alpha\beta} - v^\alpha v^\beta/v^2)$ under the switch of α and β is used.

[1] Requires further explanation if the event-averages haven't been explained so far.

To prove the first equality of eq. 3.3b, it is easiest to start from the definition of the stopping power:

$$\frac{\langle \Delta E \rangle}{\Delta x} = \frac{1}{v\Delta t}\left\langle \Delta(\sqrt{\mathbf{p}^2+m^2})\right\rangle = \frac{1}{v\Delta t}\left\langle \frac{\Delta(\mathbf{p}^2)}{2\sqrt{\mathbf{p}^2+m^2}}\right\rangle = \frac{1}{v\Delta t}\left\langle \frac{\mathbf{p}\cdot\Delta\mathbf{p}}{\sqrt{p^2+m^2}}\right\rangle = \frac{v^\alpha}{v}\frac{\Delta p^\alpha}{\Delta t}$$

$X^{\alpha\beta}$ is obtained using the event-averaged field correlators previously derived, while determining Y^α requires calculating the event-average of the correlator of one of the fields and each event's deviation of the quark-probe density from the event-averaged density, n, a quantity which is difficult to access.

In order to circumvent the challenges of calculating Y^α, it is assumed that the Fokker-Planck equation should, if the system were in equilibrium at a temperature T, admit the Boltzmann distribution function as a solution:

$$n^{\text{eq}}(\mathbf{p}) \propto \exp(E_\mathbf{p}/T), \qquad \text{with } E_\mathbf{p} = \sqrt{\mathbf{p}^2+{m_Q}^2}$$

with m_Q the mass of the quark probe. This assumption allows obtaining Y^α from $X^{\alpha\beta}$ via the simple relation:

$$Y^\alpha = \frac{v^\beta}{T}X^{\alpha\beta} \tag{3.4}$$

For the purpose of obtaining Y^α from $X^{\alpha\beta}$, T is taken to be the temperature of a thermalized QGP with an energy density equal to that of the glasma[2]. That is, the temperature which would characterize the system achieved if the glasma were allowed to reach equilibrium without expanding. [Comments on validity?] - partly discussed in sec. 5 of Mrowczyncki_2018

From eqs. 3.3 and 3.4, the final expressions for determining $\hat{q}$ and $\mathrm{d}E/\,\mathrm{d}x$ from $X^{\alpha\beta}$ become:[3]

$$\hat{q} = \frac{2}{v}\left(\delta^{\alpha\beta} - \frac{v^\alpha v^\beta}{v^2}\right)X^{\alpha\beta} \tag{3.5a}$$

$$\frac{\mathrm{d}E}{\mathrm{d}x} = -\frac{v}{T}\frac{v^\alpha v^\beta}{v^2}X^{\alpha\beta} \tag{3.5b}$$

As such, obtaining $\hat{q}$ and $\mathrm{d}E/\,\mathrm{d}x$ requires calculating the $\delta^{\alpha\beta}X^{\alpha\beta}$ and $v^\alpha v^\beta X^{\alpha\beta}$ projections of the $X^{\alpha\beta}$ tensor.

3.2 The $X^{\alpha\beta}$ tensor

The first collision term is obtained from the correlators of chromodynamic fields:[4]

$$\begin{aligned} X^{\alpha\beta}(t,\mathbf{x},\mathbf{v}) &= \frac{1}{N_c}\int_0^t \mathrm{d}t' \ \mathrm{Tr}_c\left[\left\langle F^\alpha(t,\mathbf{x})F^\beta(t',\mathbf{y})\right\rangle\right] \\ &= \frac{1}{2N_c}\int_0^t \mathrm{d}t' \left\langle F^\alpha_a(t,\mathbf{x})F^\beta_a(t',\mathbf{y})\right\rangle \end{aligned} \tag{3.6}$$

[2] The derivation of the glasma's energy density should end up before this section, so the reader will be familiar with the concept.
[3] The final form of dEdx will depend on wheteher the projection calculated will be on va*vb or va*vb/v**2
[4] The first line of this equation disagrees with [15] by a fator of $1/2$ but agrees with [28]. However, the second line is in agreement with [15]. This is presumably due to a typo in [15].

where N_c is the number of colors, Tr_c is a color trace in the fundamental representation, $\mathbf{y} = \mathbf{x} - \mathbf{v}(t - t')$, $F^\alpha \equiv F^\alpha_a t^a$[5] is the α component of the Lorentz color force, and a are color indices in the adjoint representation, which run from 1 through ${N_c}^2 - 1$. Thus, the $X^{\alpha\beta}$ component of the tensor is obtained by integrating, in time, over the full past trajectory of the probe, the correlator of the α and β components of the Lorentz color force; the first at the "present" (time t, position $\mathbf{x}$), and the second at a past point of the probe's trajectory (time t', position $\mathbf{y}$).

The Lorentz color force a quark is subject to when traveling with velocity $\mathbf{v}$ at time t and position $\mathbf{x}$ is:

$$\mathbf{F}_a(t,\mathbf{x}) = g\big(\mathbf{E}_a(t,\mathbf{x}) + \mathbf{v} \times \mathbf{B}_a(t,\mathbf{x})\big) \tag{3.7a}$$

$$F^\alpha_a(t,\mathbf{x}) = g\big(E^\alpha_a(t,\mathbf{x}) + \varepsilon^{\alpha\gamma\gamma'} v^\gamma B^{\gamma'}_a(t,\mathbf{x})\big) \tag{3.7b}$$

where g is the strong coupling constant and $\varepsilon^{\alpha\beta\gamma}$ is the Levi-Civita or antisymmetric tensor with $\varepsilon^{123} = 1$.

Plugging eq. 3.7b into eq. 3.6 allows writing the unfolded expression for $X^{\alpha\beta}$:

$$\begin{aligned} X^{\alpha\beta}(t,\mathbf{x},\mathbf{v}) = \frac{g^2}{2N_c} \int_0^t \mathrm{d}t' \Big[& \langle E^\alpha_a(t,\mathbf{x}) E^\beta_a(t-t',\mathbf{y})\rangle \\ & + \varepsilon^{\beta\gamma\gamma'} v^\gamma \left\langle E^\alpha_a(t,\mathbf{x}) B^{\gamma'}_a(t-t',\mathbf{y})\right\rangle \\ & + \varepsilon^{\alpha\gamma\gamma'} v^\gamma \left\langle B^{\gamma'}_a(t,\mathbf{x}) E^\beta_a(t-t',\mathbf{y})\right\rangle \\ & + \varepsilon^{\alpha\gamma\gamma'} \varepsilon^{\beta\delta\delta'} v^\gamma v^\delta \left\langle B^{\gamma'}_a(t,\mathbf{x}) B^{\delta'}_a(t-t',\mathbf{y})\right\rangle \Big] \end{aligned} \tag{3.8}$$

[5] A proper appendix or introduction to SU(Nc) will be necessary

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Appendix A

Derivations

A.1 Covariant Current Conservation

$$[D_\nu, J^\nu] = [D_\nu, [D_\mu, F^{\mu\nu}]] = 0 \tag{A.1}$$

where the second equality may be shown by use of the Jacobi Identity:

$$\begin{aligned} &[D_\nu, [D_\mu, F^{\mu\nu}]] = -[D_\mu, [F^{\mu\nu}, D_\nu]] - [F^{\mu\nu}, [D_\nu, D_\mu]] \\ \Leftrightarrow\ &[D_\nu, [D_\mu, F^{\mu\nu}]] = -[D_\nu, [D_\mu, F^{\mu\nu}]] - ig[F^{\mu\nu}, F_{\mu\nu}] \\ \Leftrightarrow\ &2[D_\nu, [D_\mu, F^{\mu\nu}]] = 0 \end{aligned} \tag{A.2}$$

where, in the first step, the "dummy" indices of the first term on the right are switched ($\mu \leftrightarrow \nu$) and then the antisymmetry of the anticommutator and of the field strength tensor is used; for the second step, $[D_\mu, D_\nu]$ may be shown to equal $igF_{\mu\nu}$:

$$\begin{aligned} [D_\mu, D_\nu] &= [\partial_\mu + igA_\mu, \partial_\nu + igA_\nu] = [\partial_\mu, \partial_\nu] + ig[\partial_\mu, A_\nu] + ig[A_\mu, \partial_\nu] - g^2[A^\mu, A^\nu] \\ &= 0 + ig\left(\partial_\mu A_\nu - A_\nu\partial_\mu + A_\mu\partial_\nu - \partial_\nu A_\mu + ig[A_\mu, A_\nu]\right) \\ &= ig\left((\partial_\mu A_\nu) - (\partial_\nu A_\mu) + ig[A_\mu, A_\nu]\right) = igF_{\mu\nu} \end{aligned} \tag{A.3}$$

and $[F^{\mu\nu}, F_{\mu\nu}] = 0$ because, first swapping the order in the commutator and then swapping the raised and lowered indices,

$$[F^{\mu\nu}, F_{\mu\nu}] = -[F_{\mu\nu}, F^{\mu\nu}] = -[F^{\mu\nu}, F_{\mu\nu}] \tag{A.4}$$

A.2 Equations (2.58)

A.2.1 Equation (2.58a)

From Eq. (2.48a) (left),

$$\begin{aligned} E^\eta &= -F_{+-} = -\partial_+ A_- + \partial_- A_+ - ig\,[A_+, A_-] = -\partial_+(x^+\beta) - \partial_-(x^-\beta) \\ &= -2\beta - x^+\partial_+\beta - x^-\partial_-\beta = -2\beta - \tau\partial_\tau\beta \end{aligned} \tag{A.5}$$

Taking the Fourier transforms of E^η and β,

$$\tilde{E}^\eta = -2\tilde{\beta} - \tau\partial_\tau\tilde{\beta} \tag{A.6}$$

Inserting Eq. (2.57a),

$$\tilde{E}^\eta = -2\tilde{\beta}_0\left(\frac{2J_1(\omega\tau)}{\omega\tau} + \tau\partial_\tau\left(\frac{J_1(\omega\tau)}{\omega\tau}\right)\right) = -2\tilde{\beta}_0 J_0(\omega\tau) \tag{A.7}$$

where the Bessel functions' recurrence relations were used in the last step.

Taking the limit where $\tau \to 0^+$,

$$\tilde{E}^\eta_0 \equiv \lim_{\tau\to 0^+} \tilde{E}^\eta = -2\tilde{\beta}_0 \lim_{x\to 0^+} J_0(x) = -2\tilde{\beta}_0 \tag{A.8}$$

A.2.2 Equation (2.58b)

From Eq. (2.48b) (left),

$$B^\eta = -F_{23} = -\left(\partial_2 A_3 - \partial_3 A_2 + ig[A_2, A_3]\right) = \partial_2\beta^3 - \partial_3\beta^2 - ig[\beta^2, \beta^3] \tag{A.9}$$

Two equalities must be drawn from the Coulomb gauge condition, the first for $\tilde{\beta}^i$:

$$\partial_i\beta^i = 0 \implies k_i\tilde{\beta}^i = 0 \Leftrightarrow k_i\tilde{\beta}^i_0 = 0 \tag{A.10}$$

For the second, $\tilde{\beta}^i$ is decomposed into its parallel and perpendicular components to $\mathbf{k}_\perp$:

$$\tilde{\beta}^i_0(\mathbf{k}_\perp) = k^i C_\parallel + \varepsilon^{ij}k^j C_\perp \tag{A.11}$$

where the C coefficients do not depend on $\mathbf{k}_\perp$ because no Lorentz-invariant quantity may be crafted solely from k^2 and k^3. Thus, in order for $\tilde{\beta}^i_0$ to be components of a four-vector, $C_\parallel$ and $C_\perp$ must be constants.

Applying the Coulomb gauge condition to this decomposition:

$$k_i\tilde{\beta}^i(\tau, \mathbf{k}_\perp) = 0 \implies k_i\tilde{\beta}^i_0(\mathbf{k}_\perp) = 0 \Leftrightarrow -k^2_\perp C_\parallel + 0 = 0 \implies C_\parallel = 0 \tag{A.12}$$

from which

$$[\beta_0^2, \beta_0^3] = \iint \frac{\mathrm{d}\mathbf{k}_\perp}{(2\pi)^2} \iint \frac{\mathrm{d}\mathbf{k}'_\perp}{(2\pi)^2} \left[k^3 C_\perp, -k'^2 C_\perp\right] = 0 \implies [\beta^2, \beta^3] = 0 \tag{A.13}$$

Thus, taking the Fourier transforms of B^η and β^i:

$$\tilde{B}^\eta = ik_2\tilde{\beta}^3 - ik_3\tilde{\beta}^2 = -i\varepsilon^{ij}k^i\tilde{\beta}^j \tag{A.14}$$

Taking the limit $\tau \to 0^+$ and inserting Eq. (2.57b),

$$B_0^\eta \equiv \lim_{\tau\to 0^+} B^\eta = -i\varepsilon^{ij}k^i\tilde{\beta}_0^j \lim_{\tau\to 0^+} J_0(\omega\tau) = -i\varepsilon^{ij}k^i\tilde{\beta}_0^j \tag{A.15}$$

Which may be inverted as:

$$\begin{aligned} \tilde{B}_0^\eta = -i\varepsilon^{mn}k^m\tilde{\beta}_0^n \Leftrightarrow i\varepsilon^{ij}\tilde{B}_0^\eta = \varepsilon^{ij}\varepsilon^{mn}k^m\tilde{\beta}_0^n \Leftrightarrow i\varepsilon^{ij}\tilde{B}_0^\eta = \left(\delta^{im}\delta^{jn} - \delta^{in}\delta^{jm}\right)k^m\tilde{\beta}_0^n \\ \Leftrightarrow i\varepsilon^{ij}\tilde{B}_0^\eta = k^i\tilde{\beta}_0^j - k^j\tilde{\beta}_0^i \Rightarrow i\varepsilon^{ij}k^j\tilde{B}_0^\eta = k^ik^j\tilde{\beta}_0^j - k_\perp^2\tilde{\beta}_0^j \Rightarrow \tilde{\beta}_0^i = -i\varepsilon^{ij}\frac{k^j}{k_\perp^2}\tilde{B}_0^\eta \end{aligned} \tag{A.16}$$

where, in the last step, the Coulomb gauge condition, Eq. (A.10), was used.

A.3 Equations (2.49)

In truth, the calculation of E_0^η of Eq. (2.49a) requires more than simply inserting the boundary condition for α (Eq. (2.46a)) into the expression for E^η (Eq. (A.5), for β), as the expression for E^η involves a derivative in τ, so an expression for α as $\tau \to 0$ may not be used. However, treatments of the HIC glasma problem using power expansions in τ of the α and α^i fields [29] have shown that α behaves as proportional to τ^2 as $\tau \to 0^+$, so $E_0^\eta(\mathbf{x}_\perp) = -2\alpha(\tau \to 0^+, \mathbf{x}_\perp) = -ig\delta^{ij}\left[\alpha_1^i(\mathbf{x}_\perp), \alpha_2^j(\mathbf{x}_\perp)\right]$.

As for obtaining B_0^η in Eq. (2.49b), the boundary conditions for α^i (Eq. (2.46b)) may be inserted into the expression for B^η (Eq. (A.9), for β^i), returning

$$\begin{aligned} B_0^\eta = \partial_2\alpha_1^3 - \partial_3\alpha_1^2 - ig[\alpha_1^2, \alpha_1^3] \\ + \partial_2\alpha_2^3 - \partial_3\alpha_2^2 - ig[\alpha_2^2, \alpha_2^3] \\ - ig\varepsilon^{ij}[\alpha_1^i, \alpha_2^j] \end{aligned} \tag{A.17}$$

The first and second lines are 0 because of α_1^i and α_2^i being pure gauge fields; they also correspond to $-B^\eta$ in the pre-collision quadrants. For the first line, taking $\alpha_1^i = (i/g)U^\dagger\partial^iU$:

$$\begin{aligned} &-\partial^2\alpha_1^3 + \partial^3\alpha_1^2 + ig[\alpha_1^2, \alpha_1^3] = \\ &= -\frac{i}{g}\left((\partial^2U^\dagger)\partial^3U + U^\dagger\partial^2\partial^3U - (\partial^3U^\dagger)\partial^2U - U^\dagger\partial^2\partial^3U + U^\dagger(\partial^2U)U^\dagger\partial^3U - U^\dagger(\partial^3U)U^\dagger\partial^2U\right) \\ &= 0 \end{aligned} \tag{A.18}$$

using $\partial^i(U^\dagger U) = 0 \Leftrightarrow (\partial^iU^\dagger)U = -U^\dagger(\partial^iU)$.

Appendix B

Coordinate Systems

B.1 Light-Cone Coordinates

May need to be reviewed so it has a similar structure and flow to the Bjorken coordinates appendix.

Throughout the derivations, it will be useful to work in light-cone coordinates. A four-vector $a^\mu = (a^0, a^1, a^2, a^3)$ may be expressed in light-cone coordinates as

$$a^\mu = (a^+, a^-, a^2, a^3) \equiv (a^+, a^-, \mathbf{a}_\perp), \qquad \mu = (+, -,\ 2,\ 3) \tag{B.1}$$

with

$$a^+ = \frac{a^0 + a^1}{\sqrt{2}} \qquad a^- = \frac{a^0 - a^1}{\sqrt{2}} \qquad \mathbf{a}_\perp = (a^2, a^3) \tag{B.2}$$

The dot product between two four-vectors a^μ and b^ν becomes

$$a \cdot b = a^+ b^- + a^- b^+ - \mathbf{a}_\perp \cdot \mathbf{b}_\perp = a^+ b^- + a^- b^+ - a^2 b^2 - a^3 b^3 \tag{B.3}$$

the metric being

$$g_{\mu\nu} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \tag{B.4}$$

The components of the corresponding covariant four-vector a_μ are

$$a_\mu = (a_+, a_-, a_2, a_3) = (a^-, a^+, -a^2, -a^3) \tag{B.5}$$

Considering the position four-vector,

$$x^\mu = (x^+, x^-, x^2, x^3) = \left(\frac{x^0 + x^1}{\sqrt{2}}, \frac{x^0 - x^1}{\sqrt{2}}, x^2, x^3 \right) \tag{B.6}$$

we may obtain the new base vectors. Focusing on the first two coordinates:

$$\vec{r} = x^0\vec{e}_0 + x^1\vec{e}_1, \quad \text{with} \quad \vec{e}_0 = (1,0,0,0), \quad \vec{e}_1 = (0,1,0,0) \tag{B.7}$$

$$\begin{aligned} \vec{e}_+ &= \frac{\partial \vec{r}}{\partial x^+} = \frac{1}{\sqrt{2}}\vec{e}_0 + \frac{1}{\sqrt{2}}\vec{e}_1 \\ \vec{e}_- &= \frac{\partial \vec{r}}{\partial x^-} = \frac{1}{\sqrt{2}}\vec{e}_0 - \frac{1}{\sqrt{2}}\vec{e}_1 \end{aligned} \tag{B.8}$$

Allowing us to rewrite $\vec{r}$:

$$\vec{r} = x^+\vec{e}_+ + x^-\vec{e}_- \tag{B.9}$$

Thus, the x^+ and x^- axes lie on the light cone (Fig. B.1).

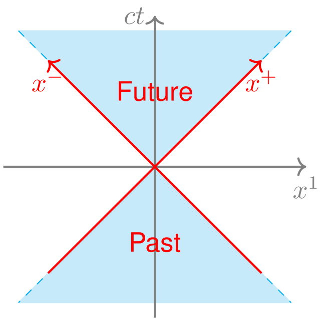

Figure B.1: Position four-vector axes in light-cone coordinates.

As for the four-gradient,

$$\partial_\mu = (\partial_+, \partial_-, \partial_2, \partial_3) = \left(\frac{\partial}{\partial x^+}, \frac{\partial}{\partial x^-}, \frac{\partial}{\partial x^2}, \frac{\partial}{\partial x^3}\right) \tag{B.10}$$

$$\partial^\mu = (\partial^+, \partial^-, \partial^2, \partial^3) = (\partial_-, \partial_+, -\partial_2, -\partial_3) \tag{B.11}$$

Abusing language and the notation, one may similarly define the Dirac matrices "in light-cone coordinates".

$$\gamma^+ = \frac{\gamma^0 + \gamma^1}{\sqrt{2}} \qquad \gamma^- = \frac{\gamma^0 - \gamma^1}{\sqrt{2}} \tag{B.12}$$

They obey anticommutation relations of the same form as the usual Dirac matrices, with the new metric of Eq. B.4.

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}, \qquad \mu, \nu = (+, -, 2, 3) \tag{B.13}$$

B.2 Bjorken/Milne Coordinates

[1] In this system, two familiar quantities are used as the first two coordinates: proper time, τ of an object with $\mathbf{x}_\perp = 0$? which set off from the origin with constant velocity? – which is invariant under boosts

[1] what Margaret calls them on page 12

along the longitudinal direction, (x^1) –, and rapidity of some body in some particular circumstances too, η. The transverse coordinates are unchanged, as in light-cone coordinates.

$$\begin{aligned} \tau &= \sqrt{(x^0)^2 - (x^1)^2} = \sqrt{2x^+x^-} \\ \eta &= \frac{1}{2}\log\left(\frac{x^0+x^1}{x^0-x^1}\right) = \frac{1}{2}\log\left(\frac{x^+}{x^-}\right) \end{aligned} \tag{B.14}$$

As for the inverse transformation,

$$x^+ = \frac{\tau}{\sqrt{2}}e^{\eta} \qquad x^- = \frac{\tau}{\sqrt{2}}e^{-\eta} \tag{B.15}$$

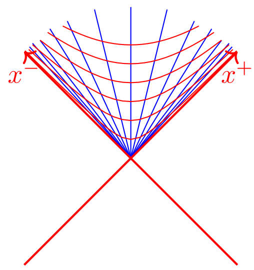

Figure B.2: Contour lines for proper time, τ, in red, and rapidity, η, in blue, in the first quadrant.

We may find the new basis vectors: versores n deviam ser "e"

$$\vec{r} = x^+\vec{e}_+ + x^-\vec{e}_-$$

$$\begin{aligned} \vec{e}_\tau &= \frac{\partial\vec{r}}{\partial\tau} = \frac{e^\eta}{\sqrt{2}}\vec{e}_+ + \frac{e^{-\eta}}{\sqrt{2}}\vec{e}_- = \frac{x^+}{\tau}\vec{e}_+ + \frac{x^-}{\tau}\vec{e}_- \\ \vec{e}_\eta &= \frac{\partial\vec{r}}{\partial\eta} = \frac{\tau}{\sqrt{2}}e^\eta\vec{e}_+ - \frac{\tau}{\sqrt{2}}e^{-\eta}\vec{e}_- = x^+\vec{e}_+ - x^-\vec{e}_- \end{aligned} \tag{B.16}$$

which are not, in general, of unit length.

By requiring

$$\vec{A} = A^+\vec{e}_+ + A^-\vec{e}_- \equiv A^\tau\vec{e}_\tau + A^\eta\vec{e}_\eta,$$

then

$$\begin{aligned} A^\tau &= \frac{x^-A^+ + x^+A^-}{\tau} \\ A^\eta &= \frac{x^-A^+ - x^+A^-}{\tau^2} \end{aligned} \tag{B.17}$$

Particularly,

$$\vec{r} = \tau\vec{e}_\tau \tag{B.18}$$

Additionally, under a boost of rapidity σ, the coordinates τ and η are changed as:

$$\tau' = \tau \tag{B.19a}$$

$$\eta' = \eta - \sigma \tag{B.19b}$$